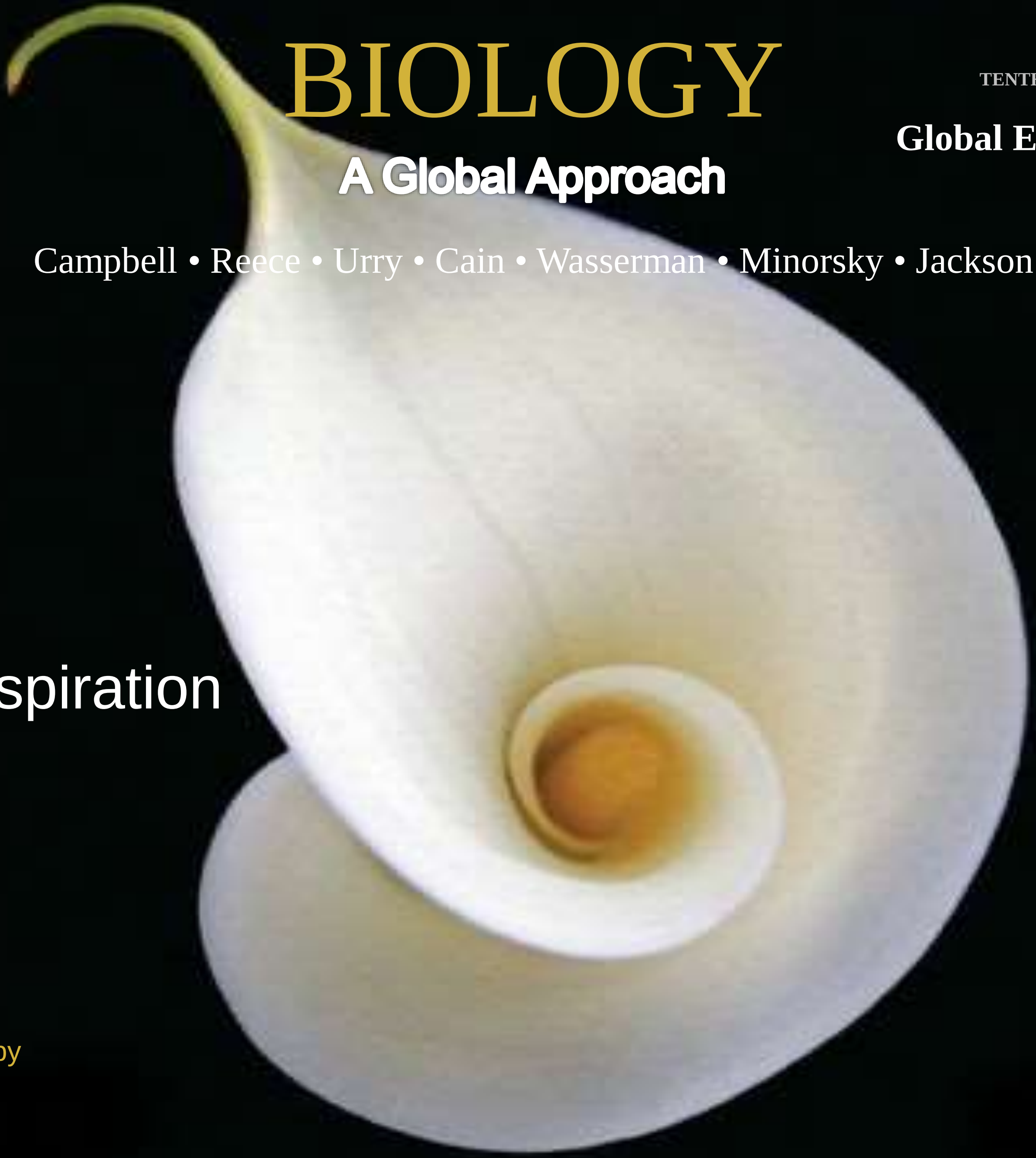
BIOLOGY

TENTH EDITION

Global Edition

A Global Approach

Campbell • Reece • Urry • Cain • Wasserman • Minorsky • Jackson



10

Cell Respiration

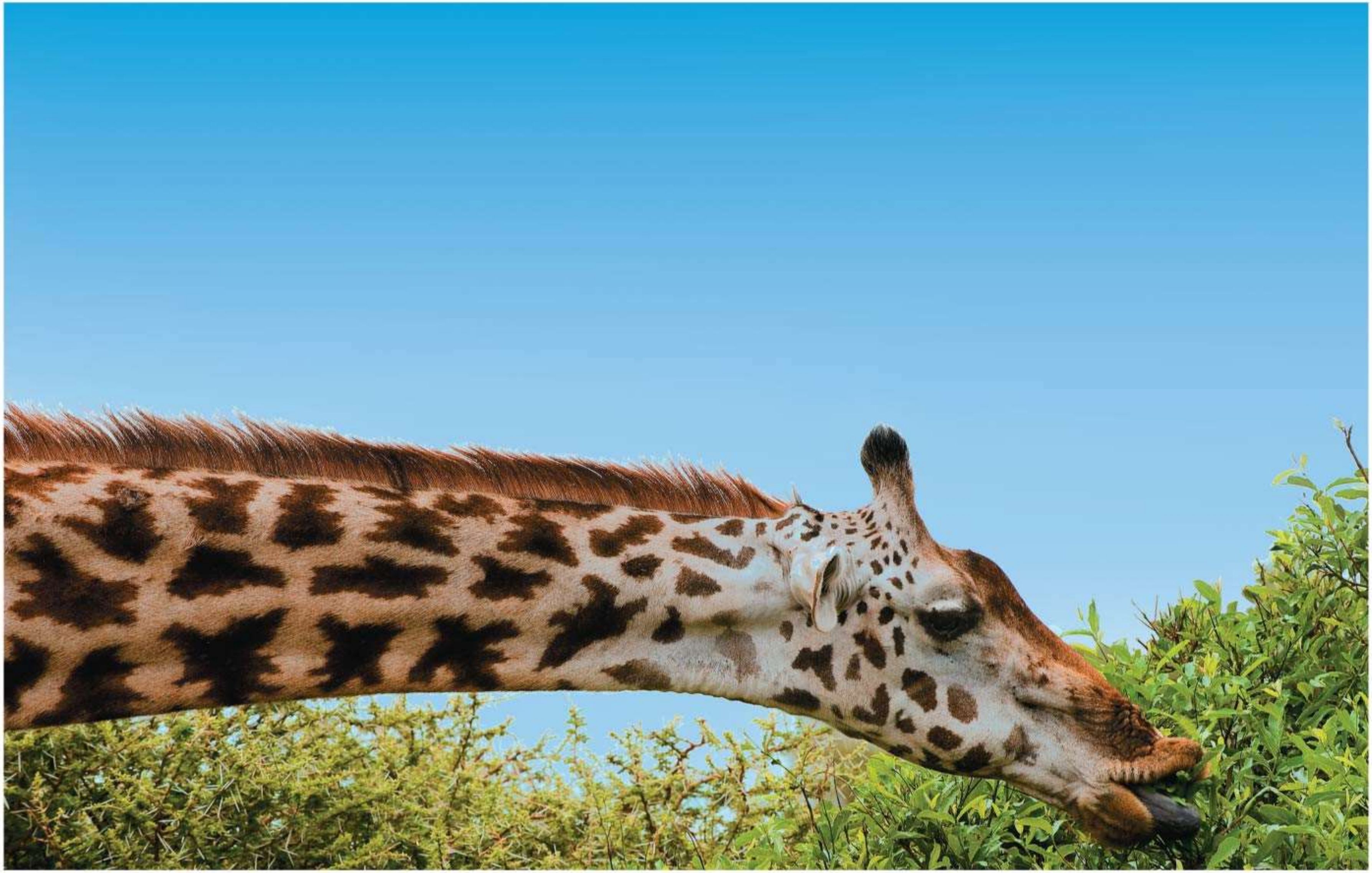
Lecture Presentation by
Nicole Tunbridge and
Kathleen Fitzpatrick

Life Is Work

a) Living cells require energy from outside sources

b) Some animals, such as the giraffe, obtain energy by eating plants, and some animals feed on other organisms that eat plants

Figure 9.1

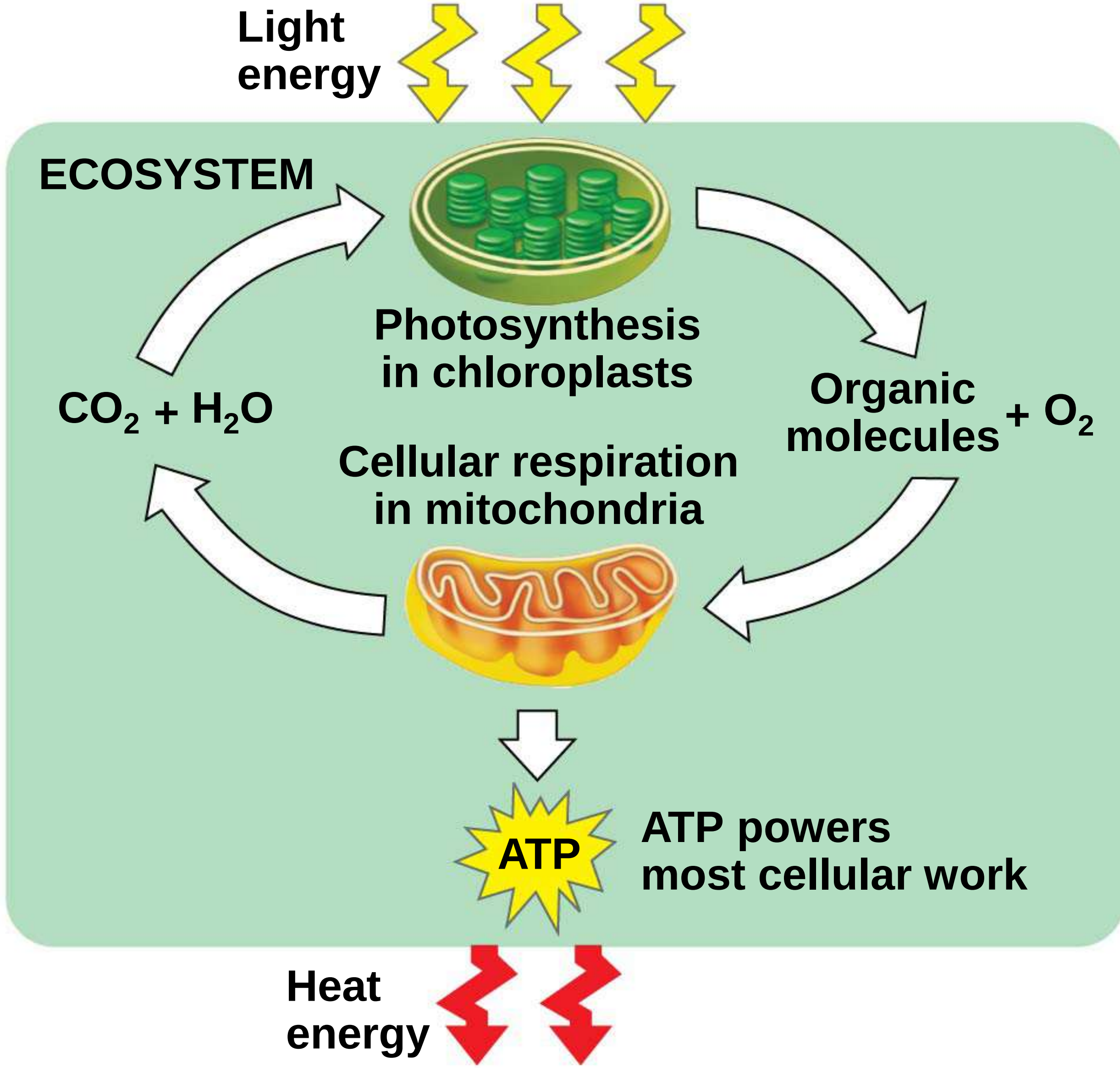


a) Energy flows into an ecosystem as sunlight and leaves as heat

b) Photosynthesis generates O_2 and organic molecules, which are used in cellular respiration

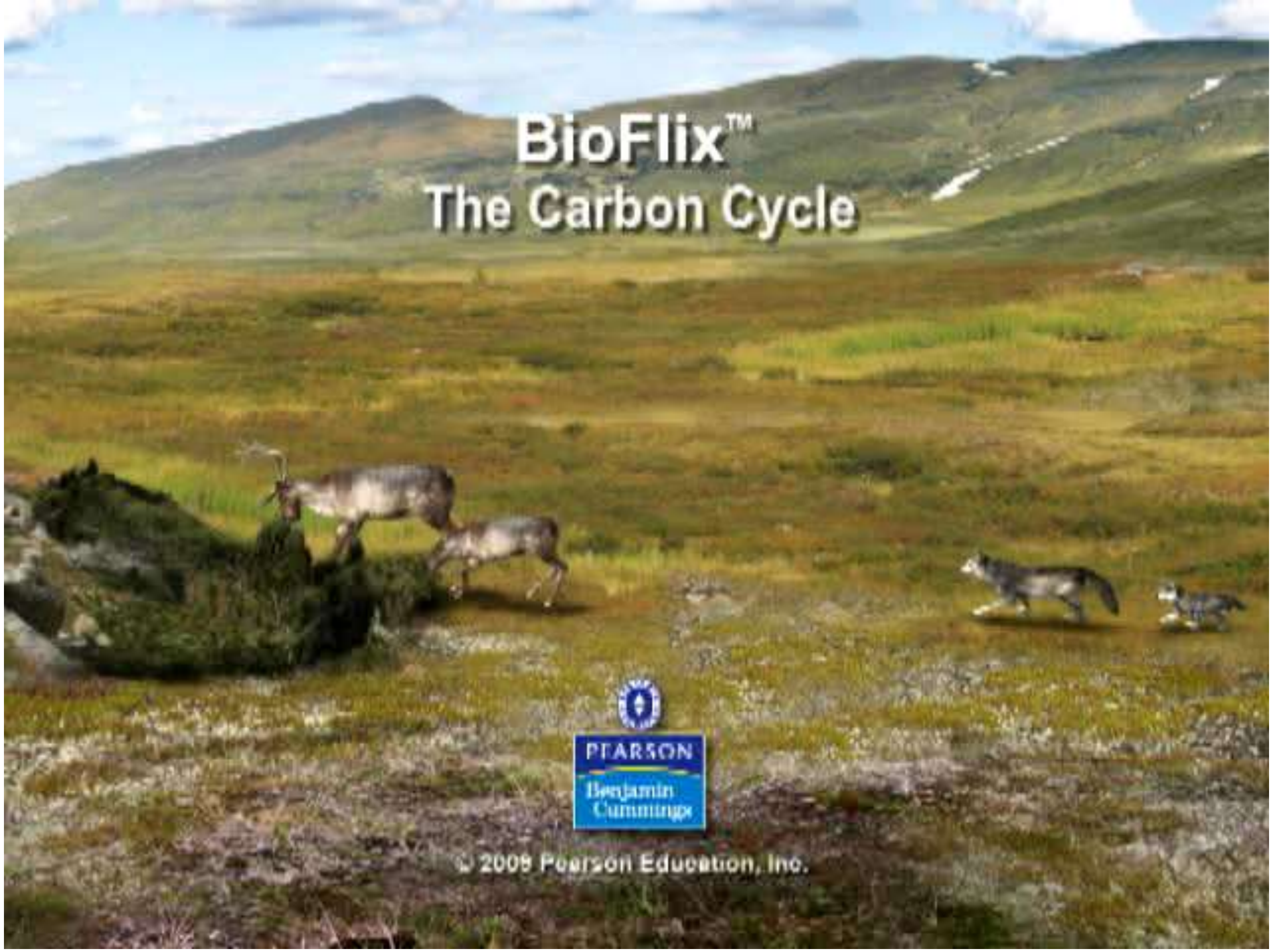
c) Cells use chemical energy stored in organic molecules to generate ATP, which powers work

Figure 9.2



Light energy
ECOSYSTEM
Photosynthesis in chloroplasts
CO_2 + H_2O
Organic molecules + O_2
Cellular respiration in mitochondria
ATP
ATP powers most cellular work
Heat energy

BioFlix: The Carbon Cycle



Concept 9.1: Catabolic pathways yield energy by oxidizing organic fuels

a) Catabolic pathways release stored energy by breaking down complex molecules

b) Electron transfer plays a major role in these pathways

c) These processes are central to cellular respiration

Catabolic Pathways and Production of ATP

a) The breakdown of organic molecules is exergonic

b) Fermentation is a partial degradation of sugars that occurs without O_2

c) Aerobic respiration consumes organic molecules and O_2 and yields ATP

d) Anaerobic respiration is similar to aerobic respiration but consumes compounds other than O_2

a) **Cellular respiration** includes both aerobic and anaerobic respiration but is often used to refer to aerobic respiration

b) Although carbohydrates, fats, and proteins are all consumed as fuel, it is helpful to trace cellular respiration with the sugar glucose

$$C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O + \text{Energy (ATP + heat)}$$

Redox Reactions: Oxidation and Reduction

a) The transfer of electrons during chemical reactions releases energy stored in organic molecules

b) This released energy is ultimately used to synthesize ATP

The Principle of Redox

a) Chemical reactions that transfer electrons between reactants are called oxidation-reduction reactions, or **redox reactions**

b) In **oxidation**, a substance loses electrons, or is oxidized

c) In **reduction**, a substance gains electrons, or is reduced (the amount of positive charge is reduced)

Figure 9.UN01

becomes oxidized
(loses electron)

$$Na + Cl \longrightarrow Na^+ + Cl^-$$

becomes reduced
(gains electron)

Figure 9.UN02

becomes oxidized

$$Xe^- + Y \longrightarrow X + Ye^-$$

becomes reduced

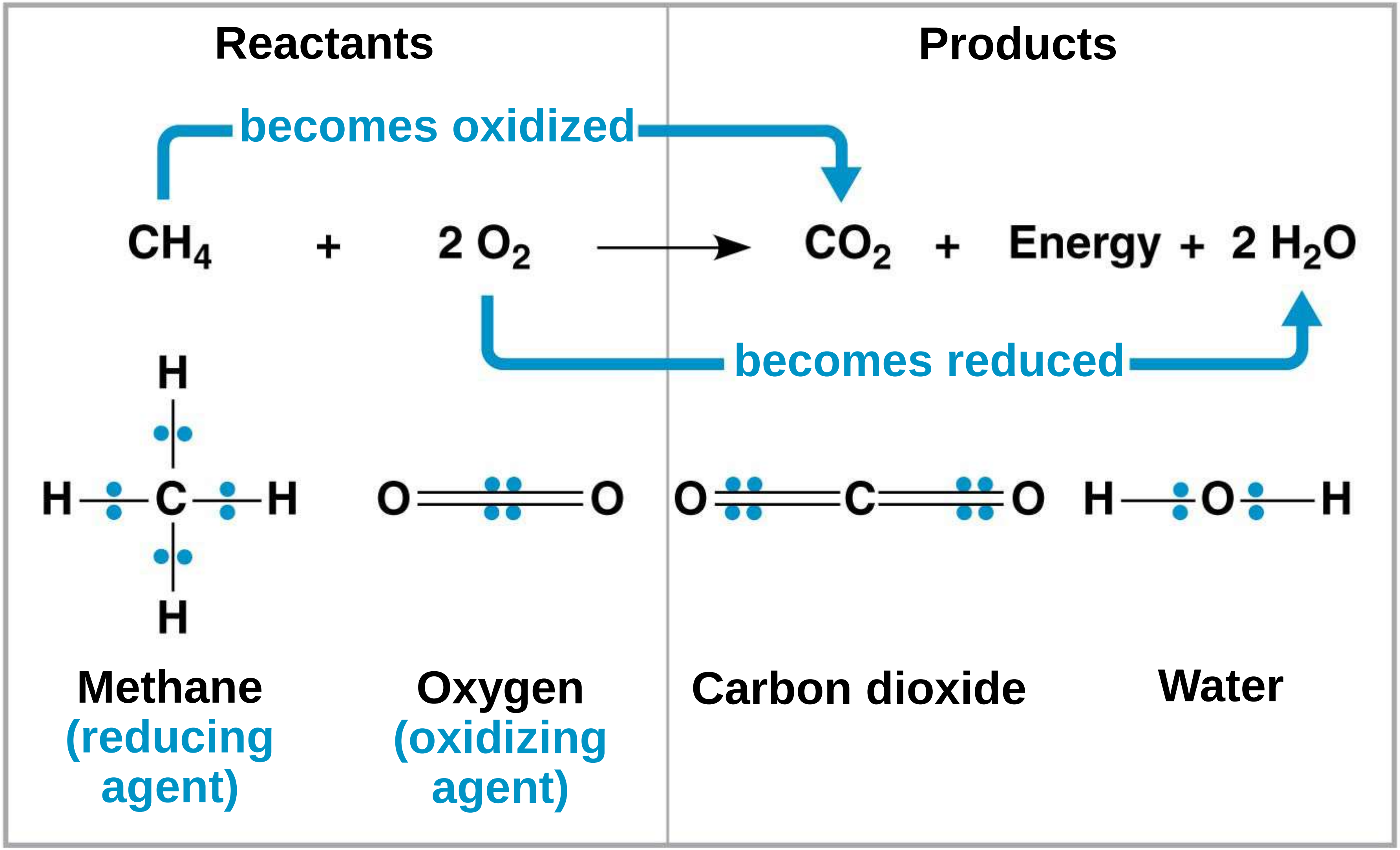
a) The electron donor is called the **reducing agent**

b) The electron receptor is called the **oxidizing agent**

c) Some redox reactions do not transfer electrons but change the electron sharing in covalent bonds

d) An example is the reaction between methane and O_2

Figure 9.3


Reactants
Products
becomes oxidized
CH4 + 2 O2 → CO2 + Energy + 2 H2O
becomes reduced
Methane
(reducing agent)
Oxygen
(oxidizing agent)
Carbon dioxide
Water

Oxidation of Organic Fuel Molecules During Cellular Respiration

a) During cellular respiration, the fuel (such as glucose) is oxidized, and O_2 is reduced

Figure 9.UN03

becomes oxidized

$$C_6H_{12}O_6 + 6\ O_2 \longrightarrow 6\ CO_2 + 6\ H_2O + \text{Energy}$$

becomes reduced

Stepwise Energy Harvest via NAD^+ and the Electron Transport Chain

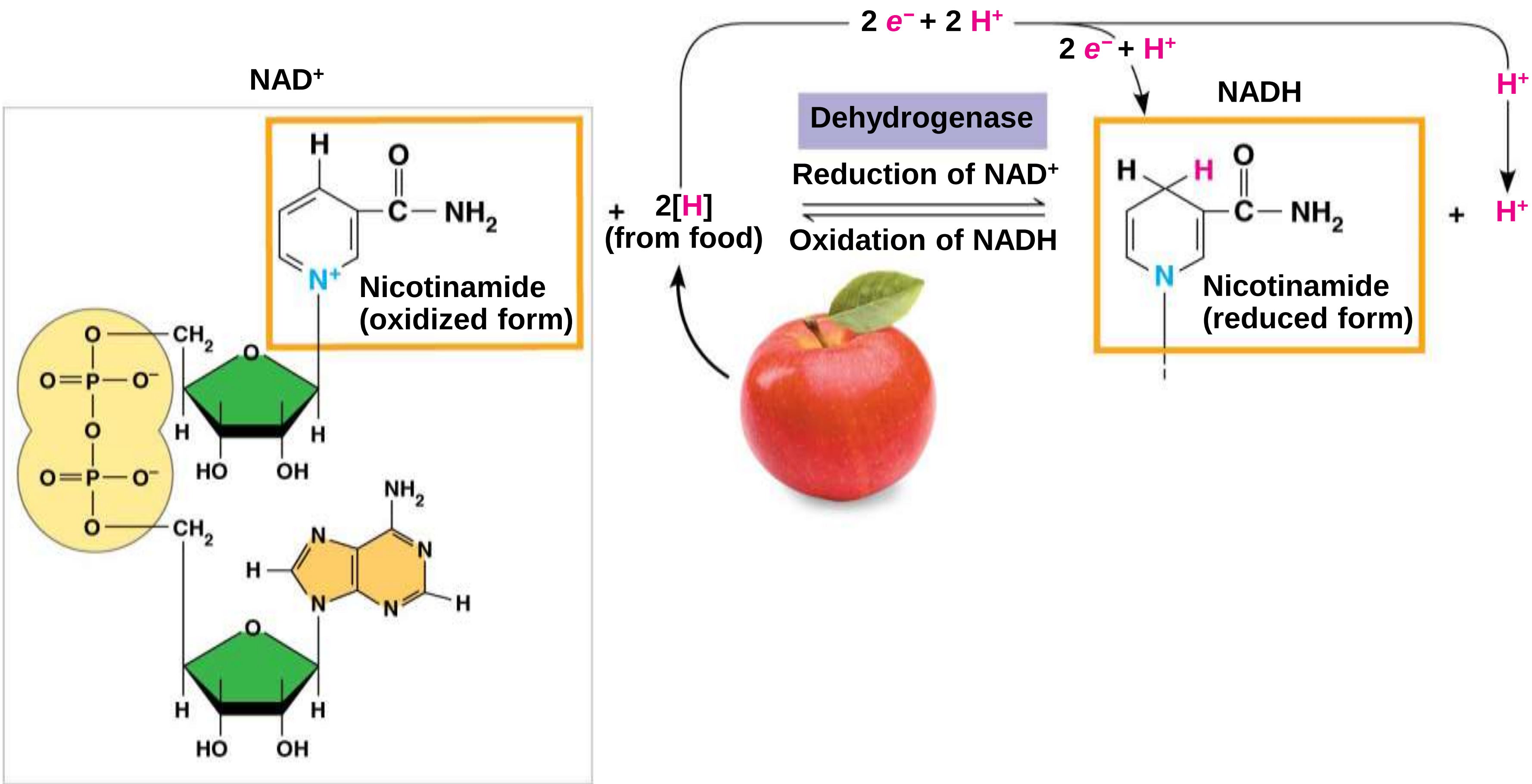
a) In cellular respiration, glucose and other organic molecules are broken down in a series of steps

b) Electrons from organic compounds are usually first transferred to **NAD^+**, a coenzyme

c) As an electron acceptor, NAD^+ functions as an oxidizing agent during cellular respiration

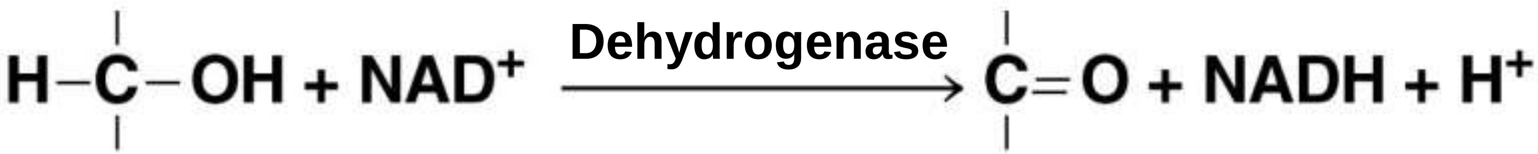
d) Each NADH (the reduced form of NAD^+) represents stored energy that is tapped to synthesize ATP

Figure 9.4



NAD+
Nicotinamide (oxidized form)
Dehydrogenase
Reduction of NAD+
Oxidation of NADH
+ 2[H] (from food)
2 e− + 2 H+
2 e− + H+
NADH
Nicotinamide (reduced form)
+ H+
H+

Figure 9.UN04



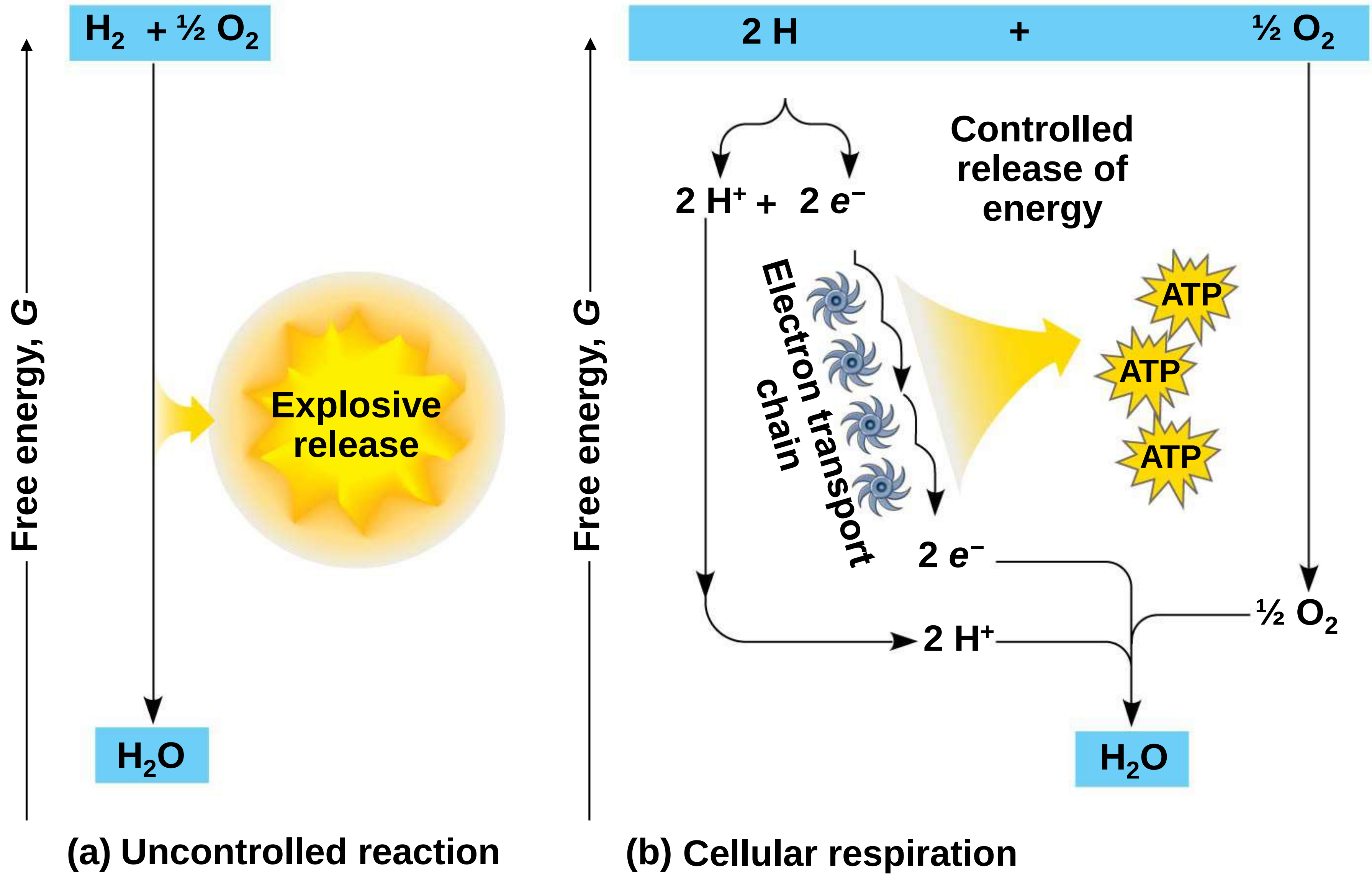
a) NADH passes the electrons to the **electron transport chain**

b) Unlike an uncontrolled reaction, the electron transport chain passes electrons in a series of steps instead of one explosive reaction

c) O_2 pulls electrons down the chain in an energy-yielding tumble

d) The energy yielded is used to regenerate ATP

Figure 9.5



H_2 + ½ O_2
Free energy, G
Explosive release
H_2O
(a) Uncontrolled reaction
2 H
+
½ O_2
Free energy, G
$2 H^+$ + $2 e^-$
Controlled release of energy
Electron transport chain
ATP
ATP
ATP
$2 e^-$
½ O_2
$2 H^+$
H_2O
(b) Cellular respiration

The Stages of Cellular Respiration: *A Preview*

a) Harvesting of energy from glucose has three stages

   a) **Glycolysis** (breaks down glucose into two molecules of pyruvate)

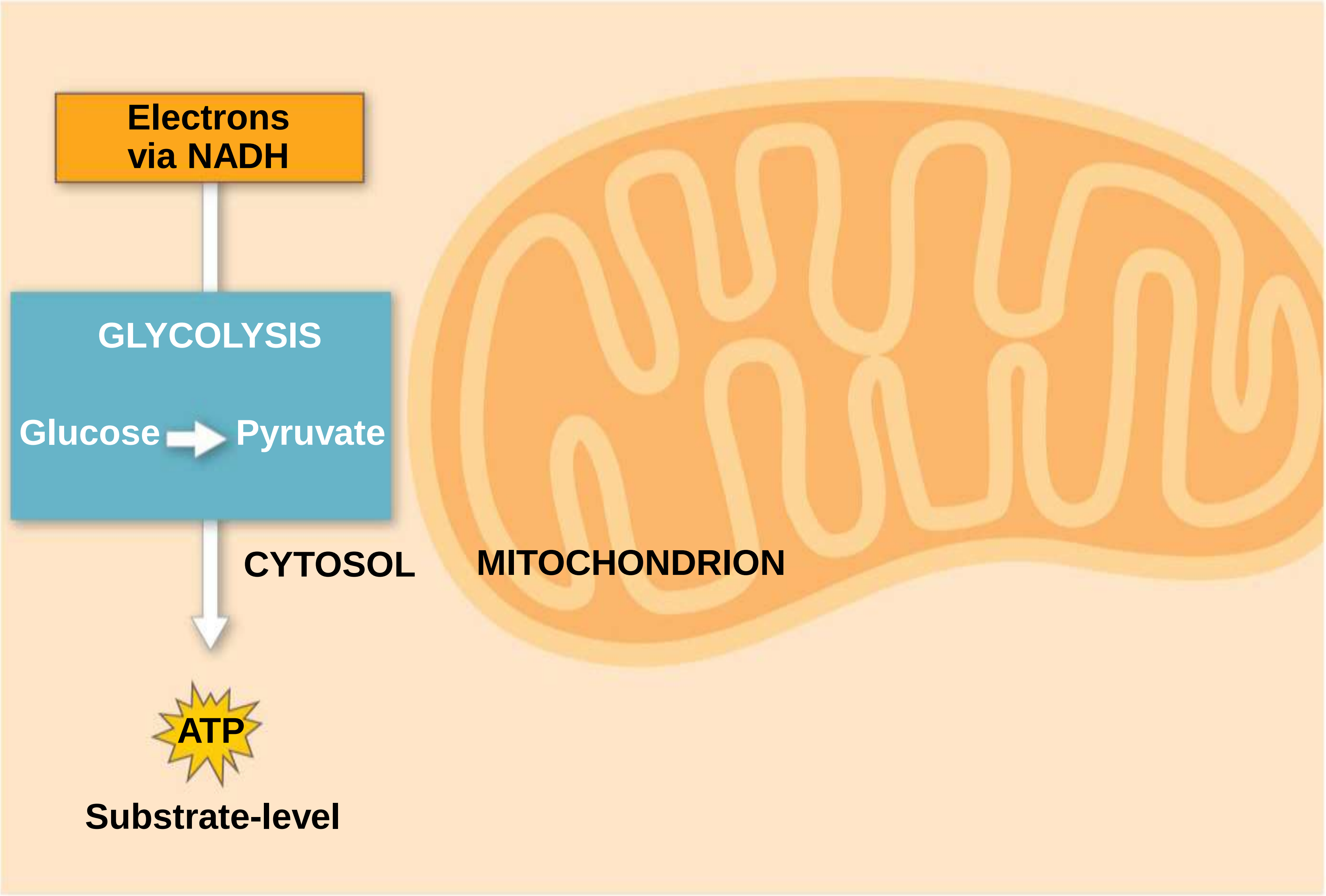
   b) The **citric acid cycle** (completes the breakdown of glucose)

   c) **Oxidative phosphorylation** (accounts for most of the ATP synthesis)

Figure 9.UN05

1. GLYCOLYSIS (color-coded blue throughout the chapter)
2. PYRUVATE OXIDATION and the CITRIC ACID CYCLE (color-coded orange)
3. OXIDATIVE PHOSPHORYLATION: Electron transport and chemiosmosis (color-coded purple)

Figure 9.6-1



Electrons via NADH
GLYCOLYSIS
Glucose
Pyruvate
CYTOSOL
MITOCHONDRION
ATP
Substrate-level

Figure 9.6-2

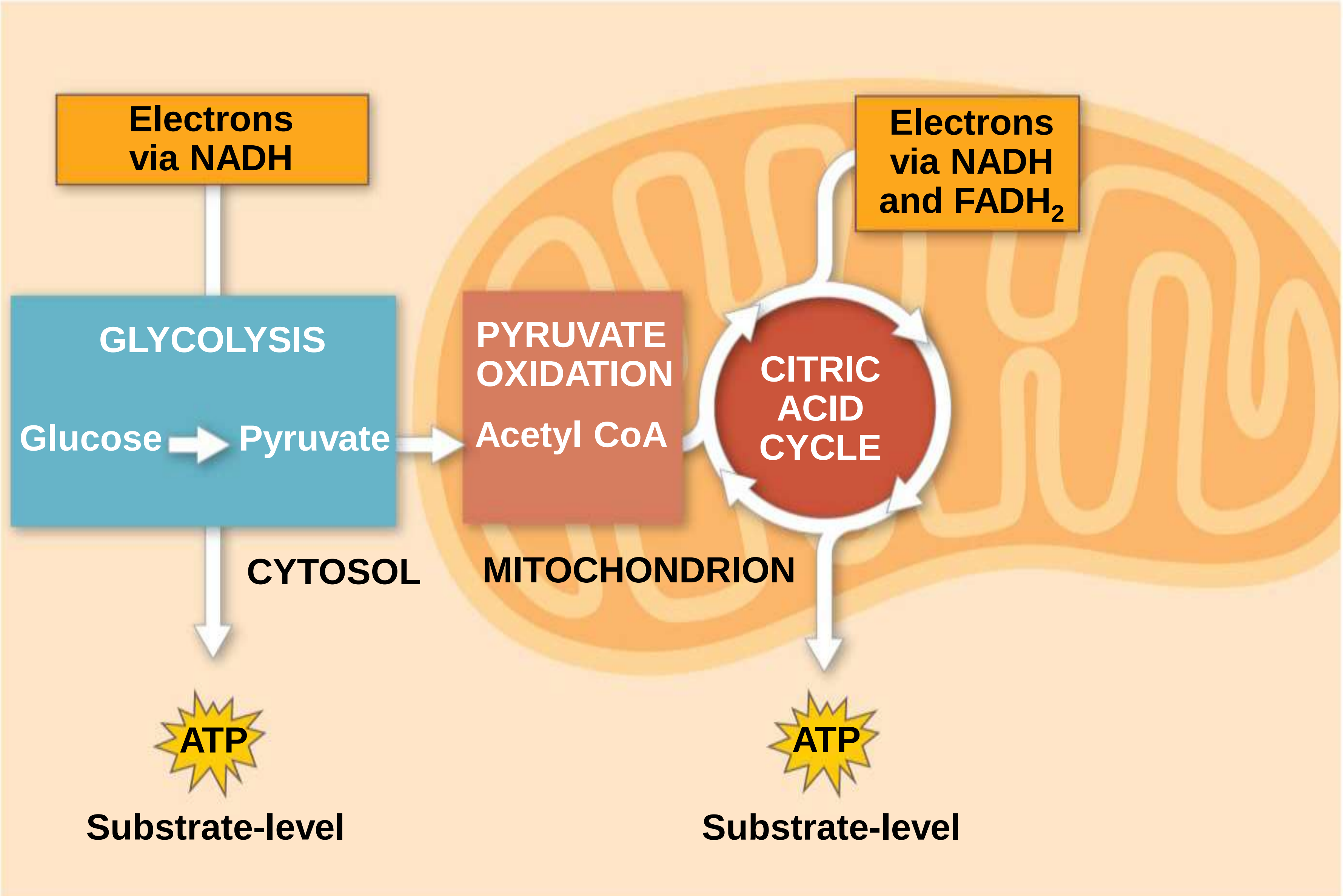
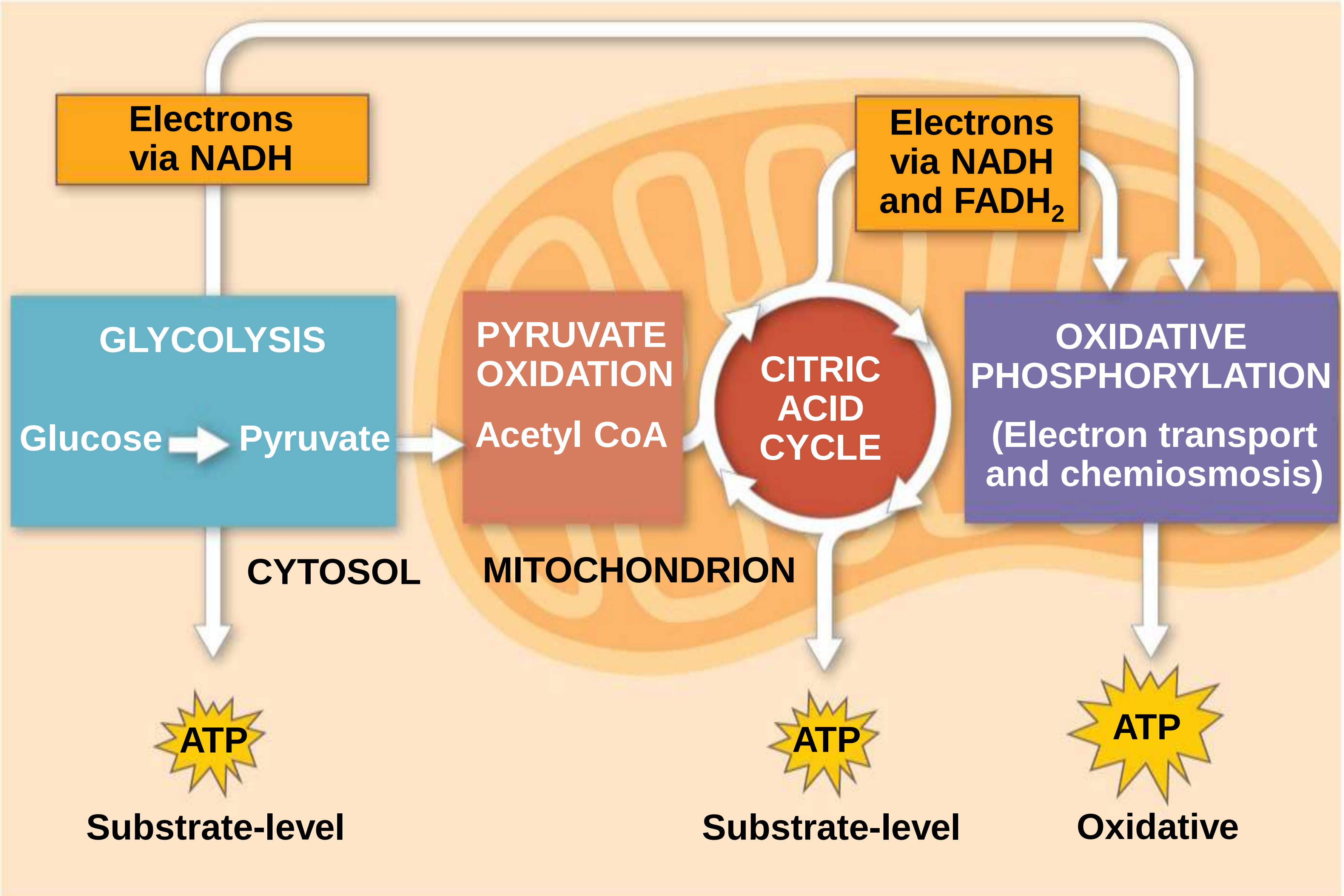

Electrons
via NADH
Electrons
via NADH
and FADH$_2$
GLYCOLYSIS
Glucose
Pyruvate
PYRUVATE
OXIDATION
Acetyl CoA
CITRIC
ACID
CYCLE
CYTOSOL
MITOCHONDRION
ATP
ATP
Substrate-level
Substrate-level

Figure 9.6-3



Electrons via NADH
Electrons via NADH and FADH$_2$
GLYCOLYSIS
Glucose
Pyruvate
PYRUVATE OXIDATION
Acetyl CoA
CITRIC ACID CYCLE
OXIDATIVE PHOSPHORYLATION
(Electron transport and chemiosmosis)
CYTOSOL
MITOCHONDRION
ATP
ATP
ATP
Substrate-level
Substrate-level
Oxidative

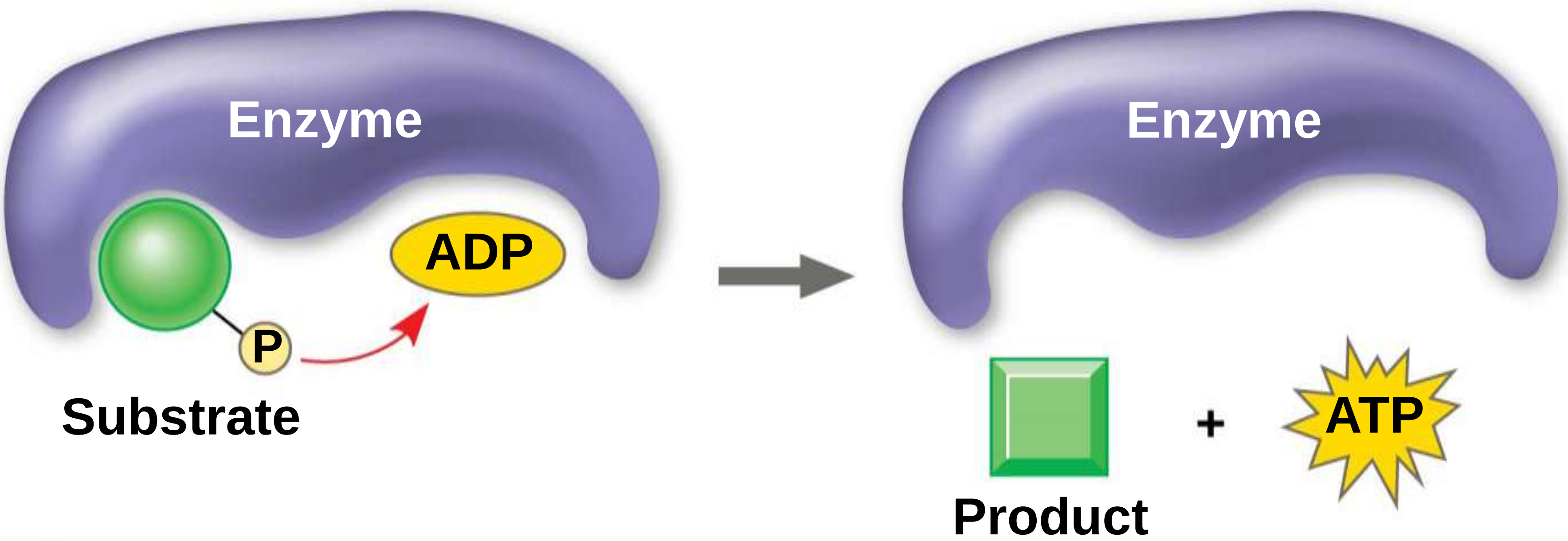
a) The process that generates most of the ATP is called **oxidative phosphorylation** because it is powered by redox reactions

a) Oxidative phosphorylation accounts for almost 90% of the ATP generated by cellular respiration

b) A smaller amount of ATP is formed in glycolysis and the citric acid cycle by **substrate-level phosphorylation**

c) For each molecule of glucose degraded to CO_2 and water by respiration, the cell makes up to 32 molecules of ATP

Figure 9.7



Concept 10.2: Glycolysis harvests chemical energy by oxidizing glucose to pyruvate

a) Glycolysis (“sugar splitting”) breaks down glucose into two molecules of pyruvate

b) Glycolysis occurs in the cytoplasm and has two major phases

   a) Energy investment phase

   b) Energy payoff phase

c) Glycolysis occurs whether or not O_2 is present

Figure 9.UN06

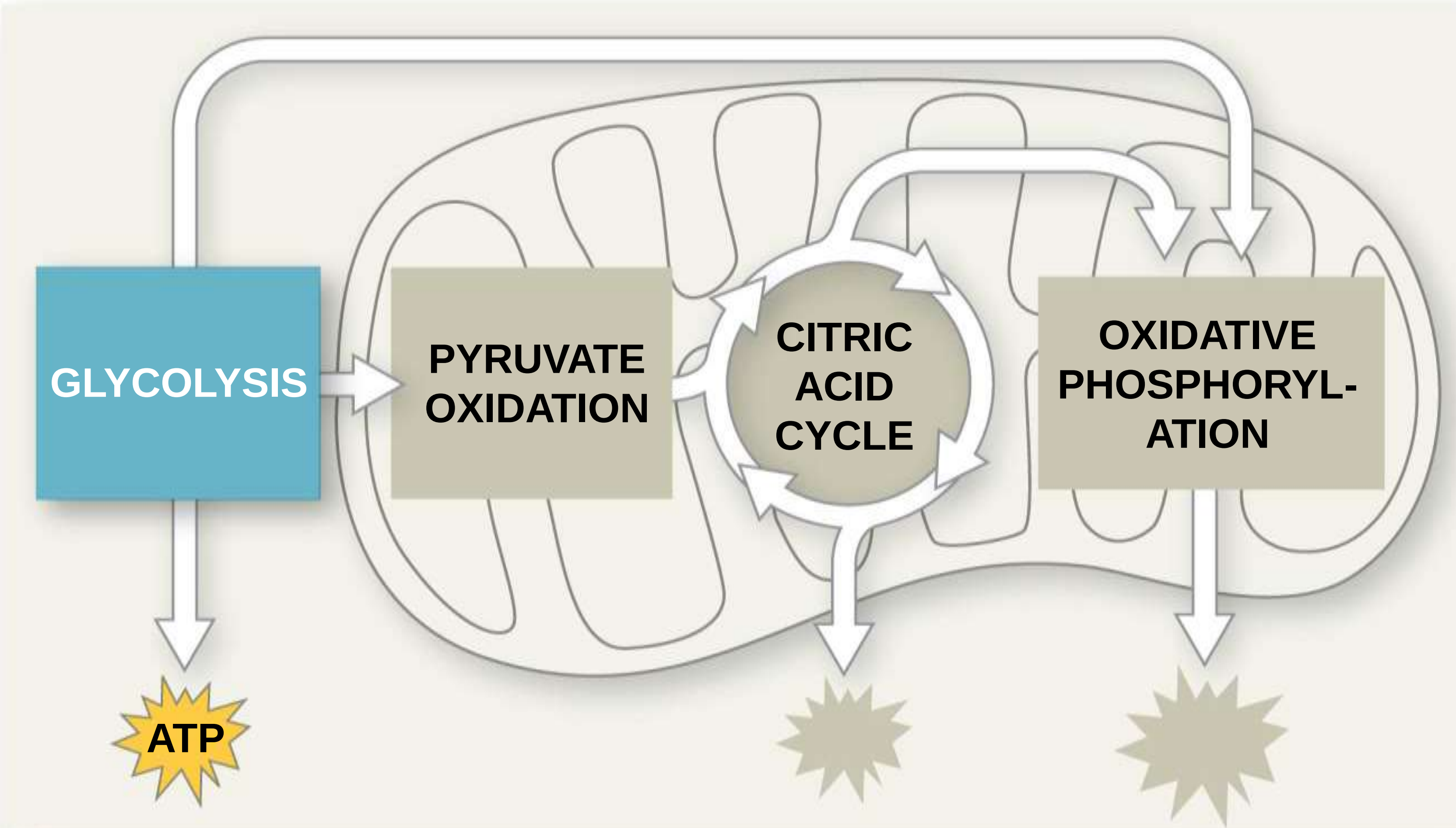


Figure 9.8

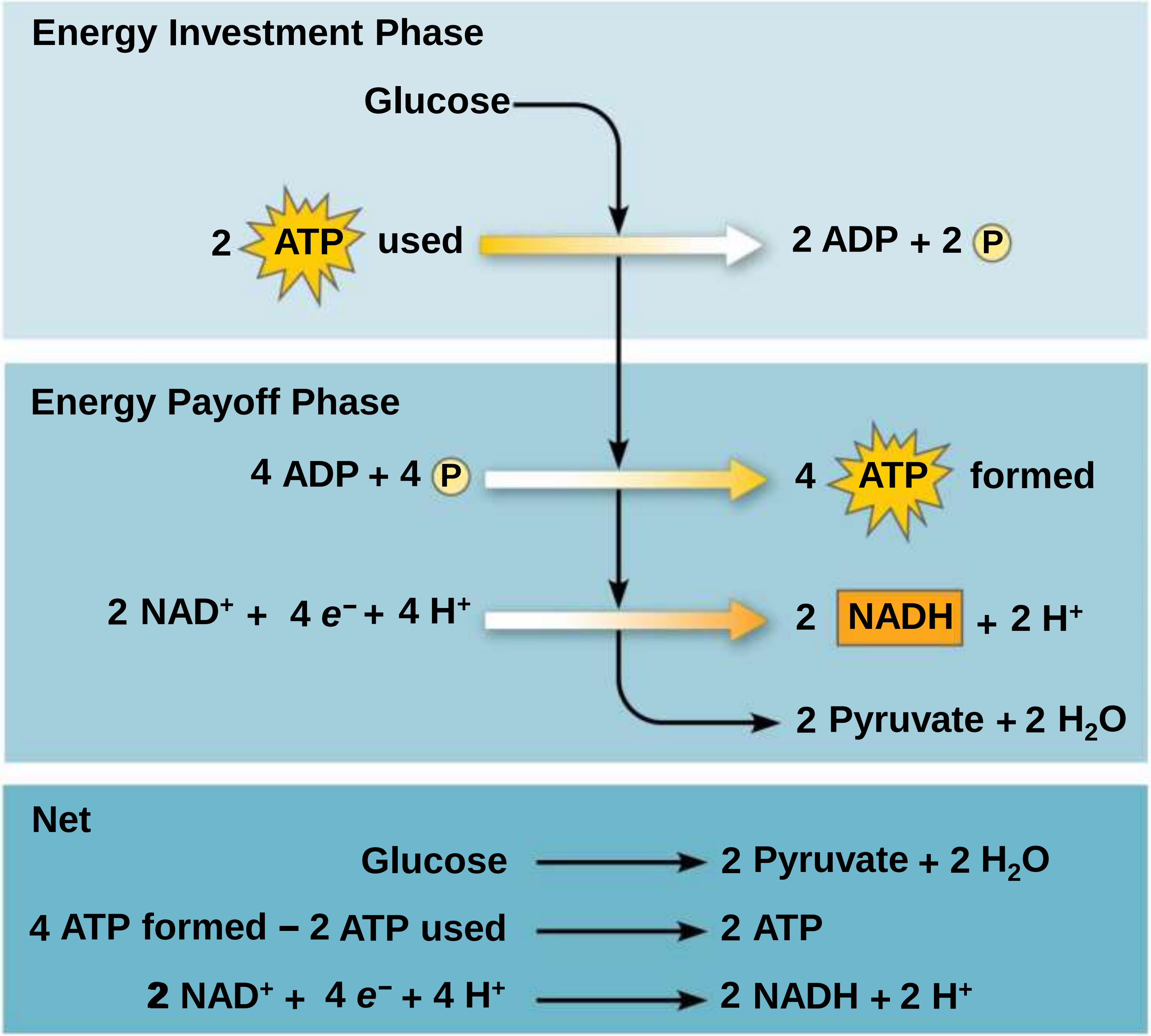

Energy Investment Phase
Glucose
2 ATP used
2 ADP + 2 P
Energy Payoff Phase
4 ADP + 4 P
4 ATP formed
2 NAD$^+$ + 4 e^- + 4 H$^+$
2 NADH + 2 H$^+$
2 Pyruvate + 2 H_2O
Net
Glucose
2 Pyruvate + 2 H_2O
4 ATP formed – 2 ATP used
2 ATP
2 NAD$^+$ + 4 e^- + 4 H$^+$
2 NADH + 2 H$^+$

Figure 9.9aa-3

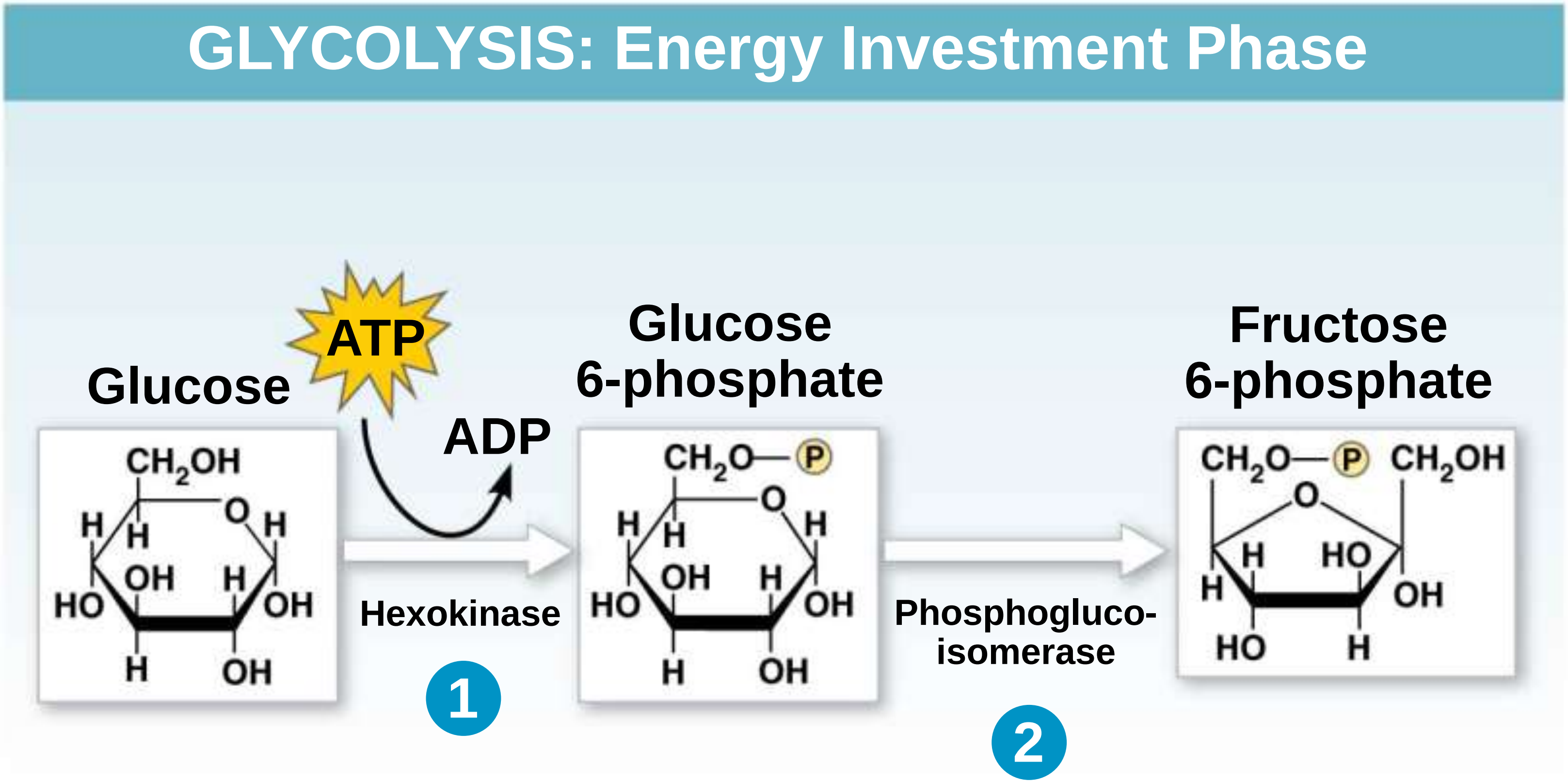


Figure 9.9ab-3

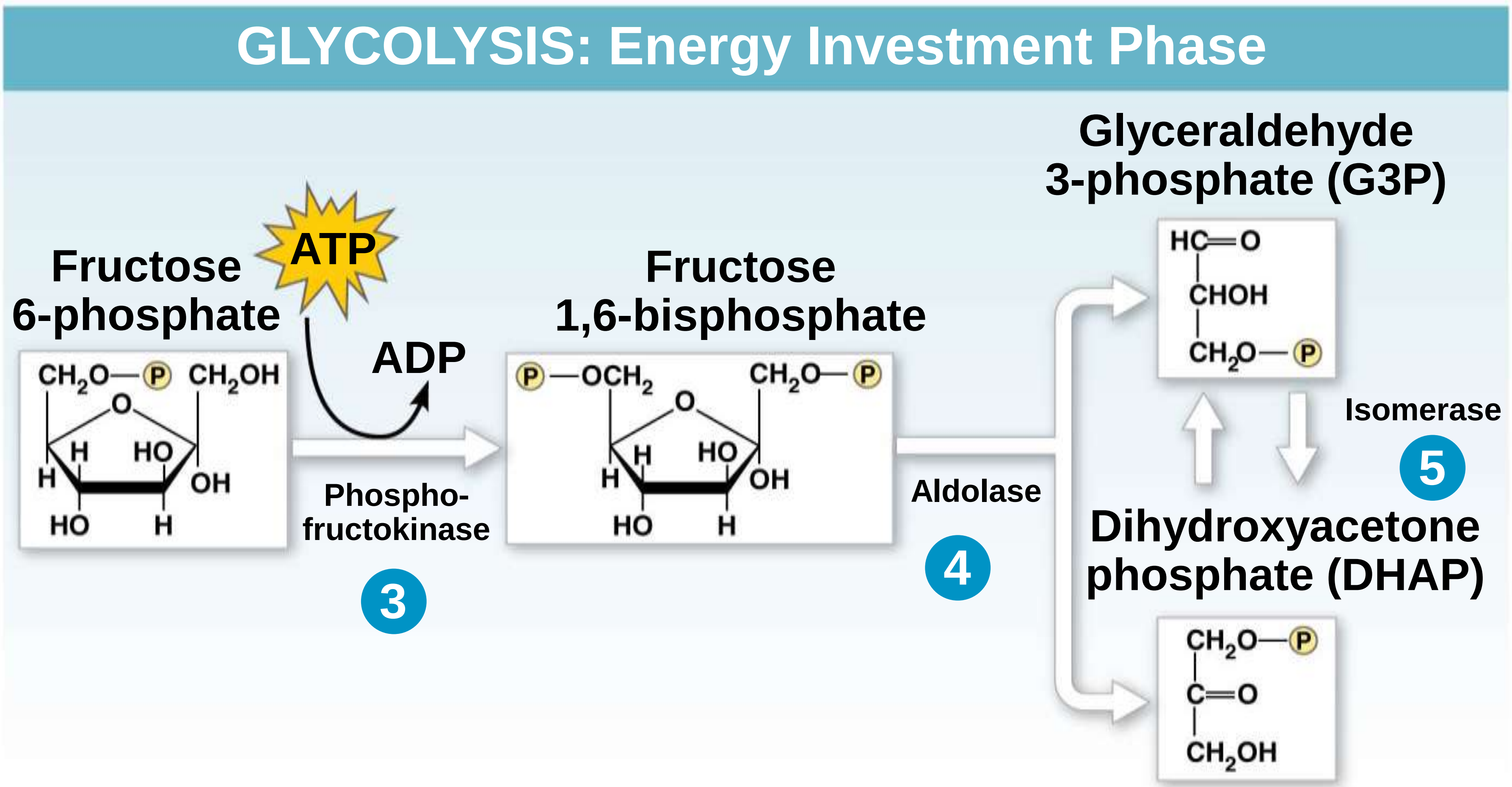


Figure 9.9ba-3

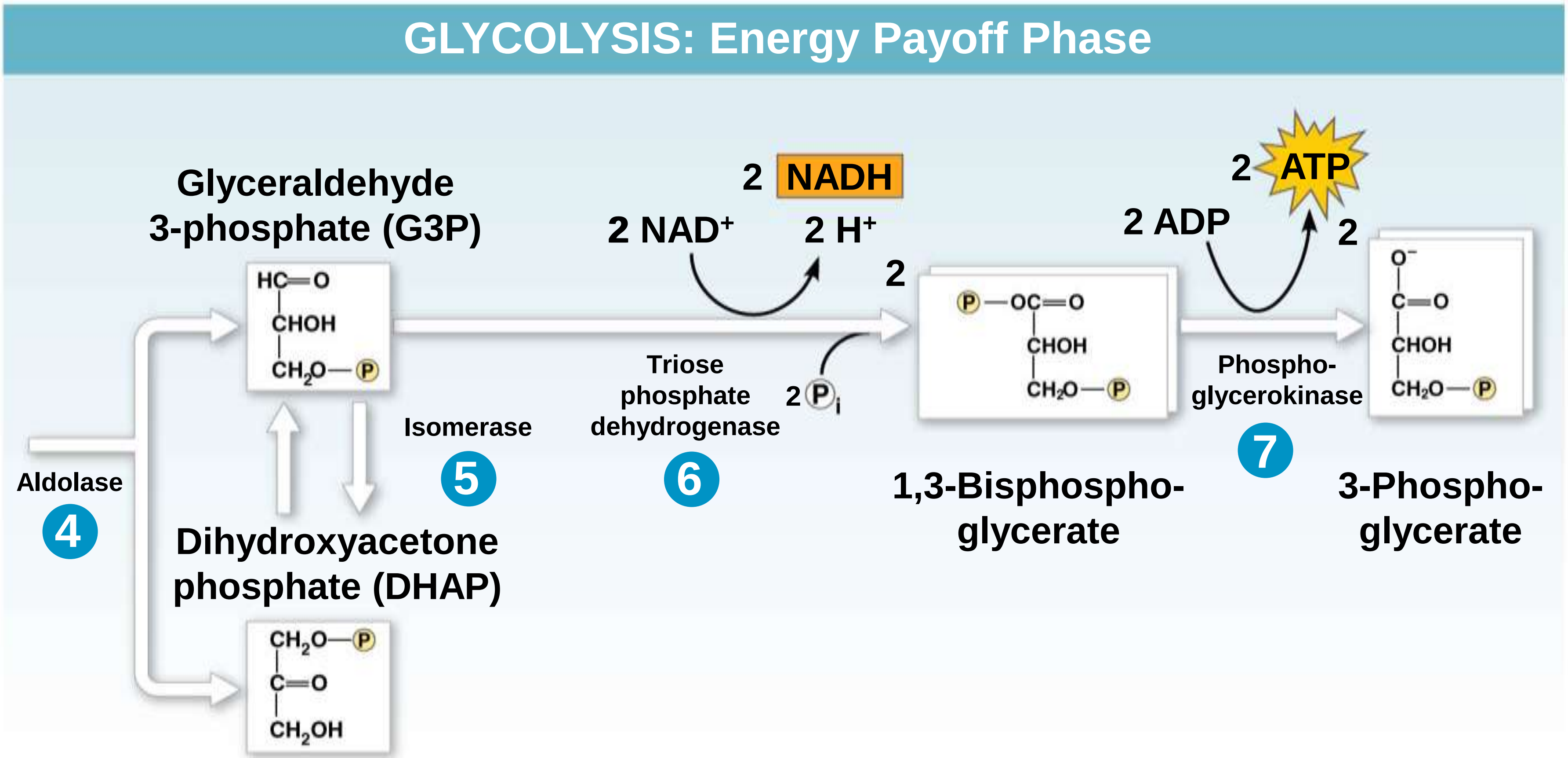
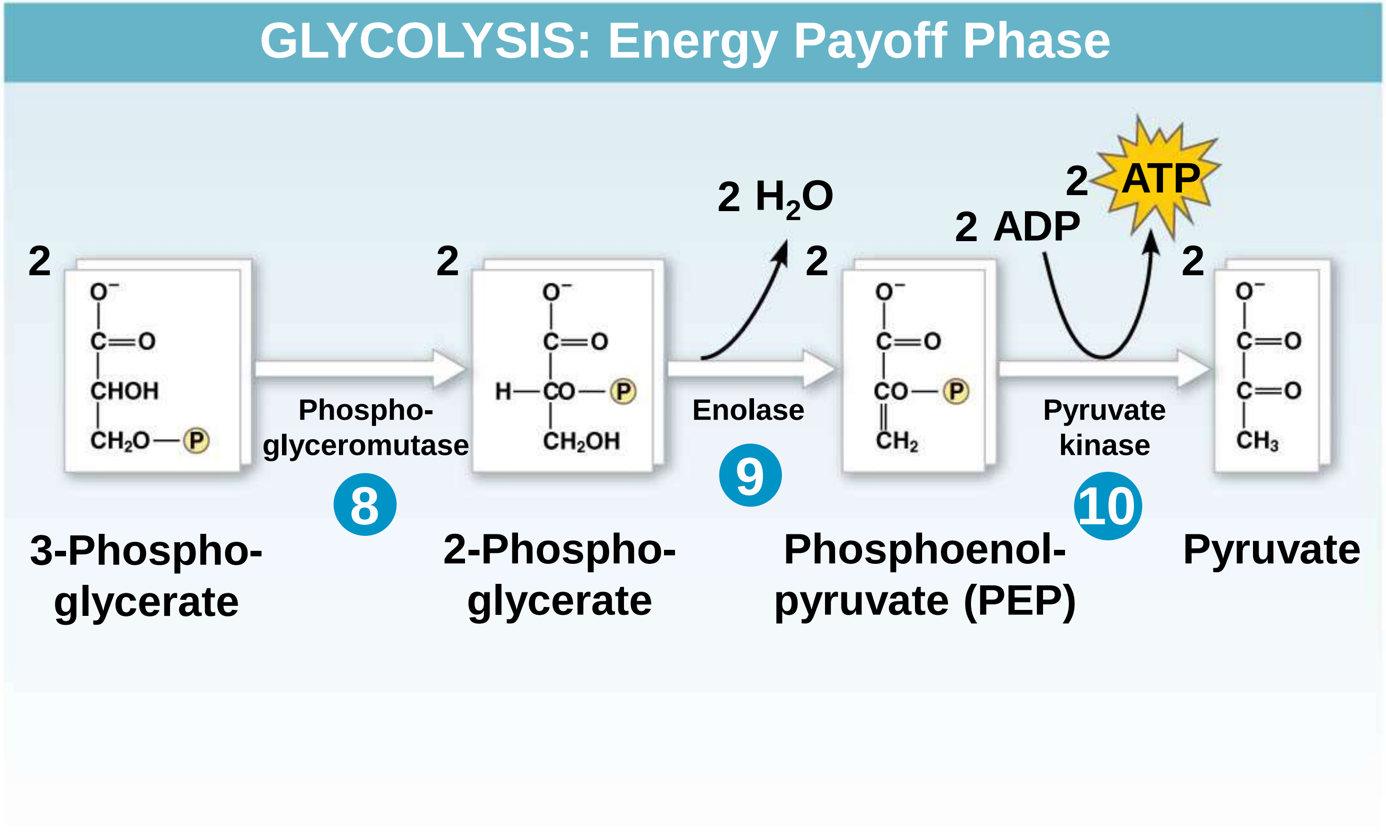


Figure 9.9bb-3



Concept 9.3: After pyruvate is oxidized, the citric acid cycle completes the energy-yielding oxidation of organic molecules

a) In the presence of O_2, pyruvate enters the mitochondrion (in eukaryotic cells) where the oxidation of glucose is completed

Oxidation of Pyruvate to Acetyl CoA

a) Before the citric acid cycle can begin, pyruvate must be converted to acetyl Coenzyme A (**acetyl CoA**), which links glycolysis to the citric acid cycle

b) This step is carried out by a multienzyme complex that catalyses three reactions

Figure 9.UN07

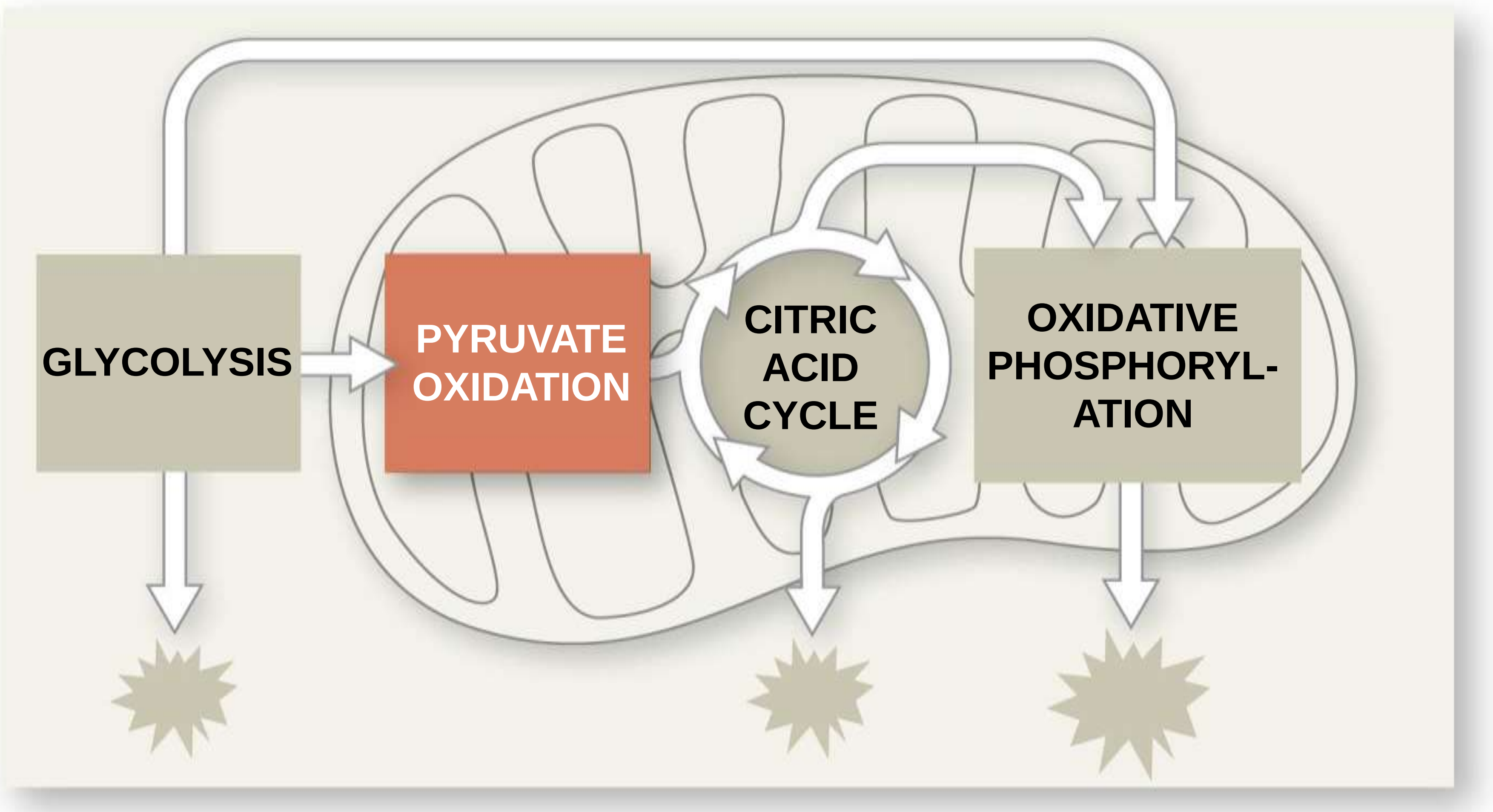
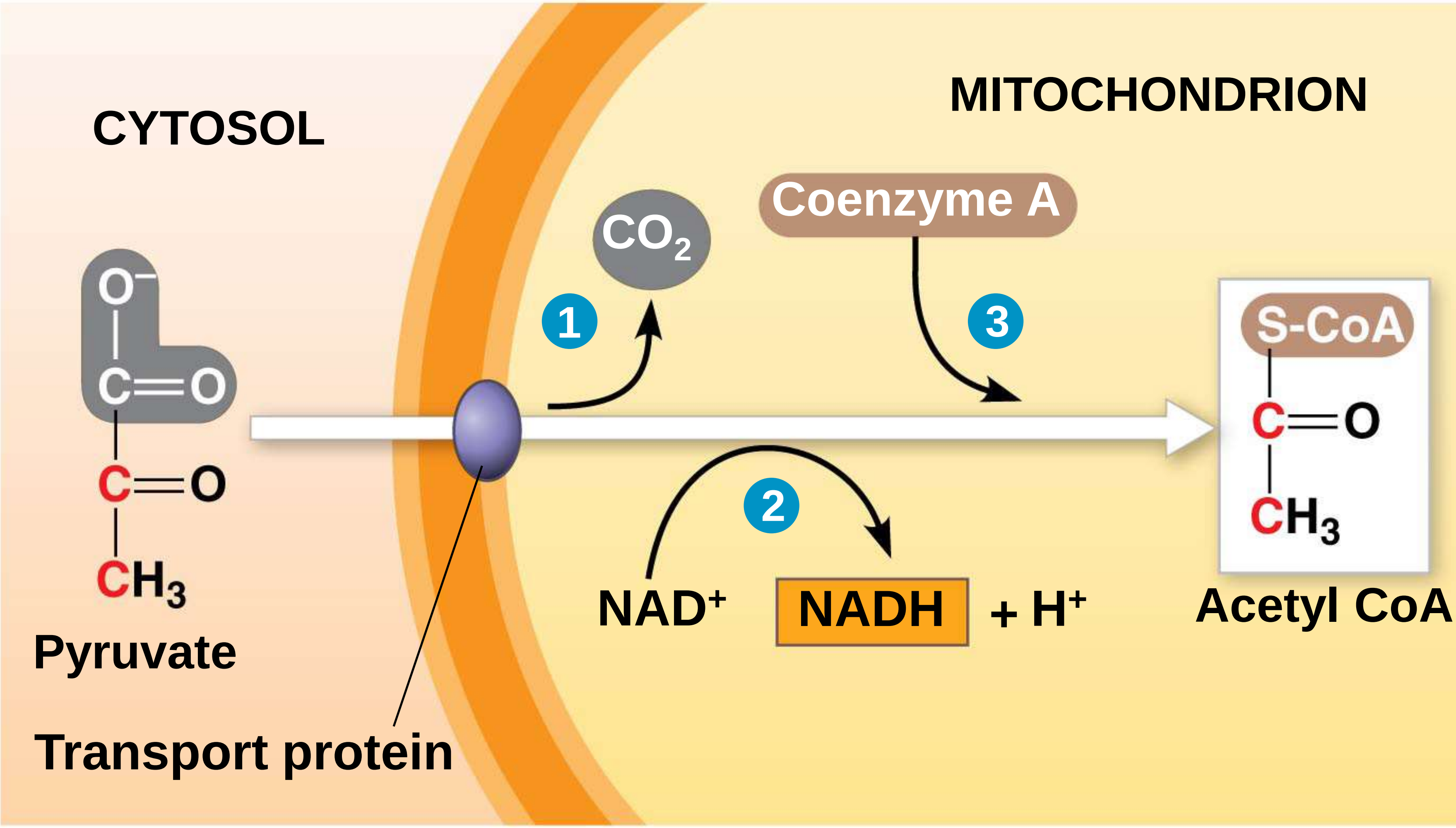


Figure 9.10

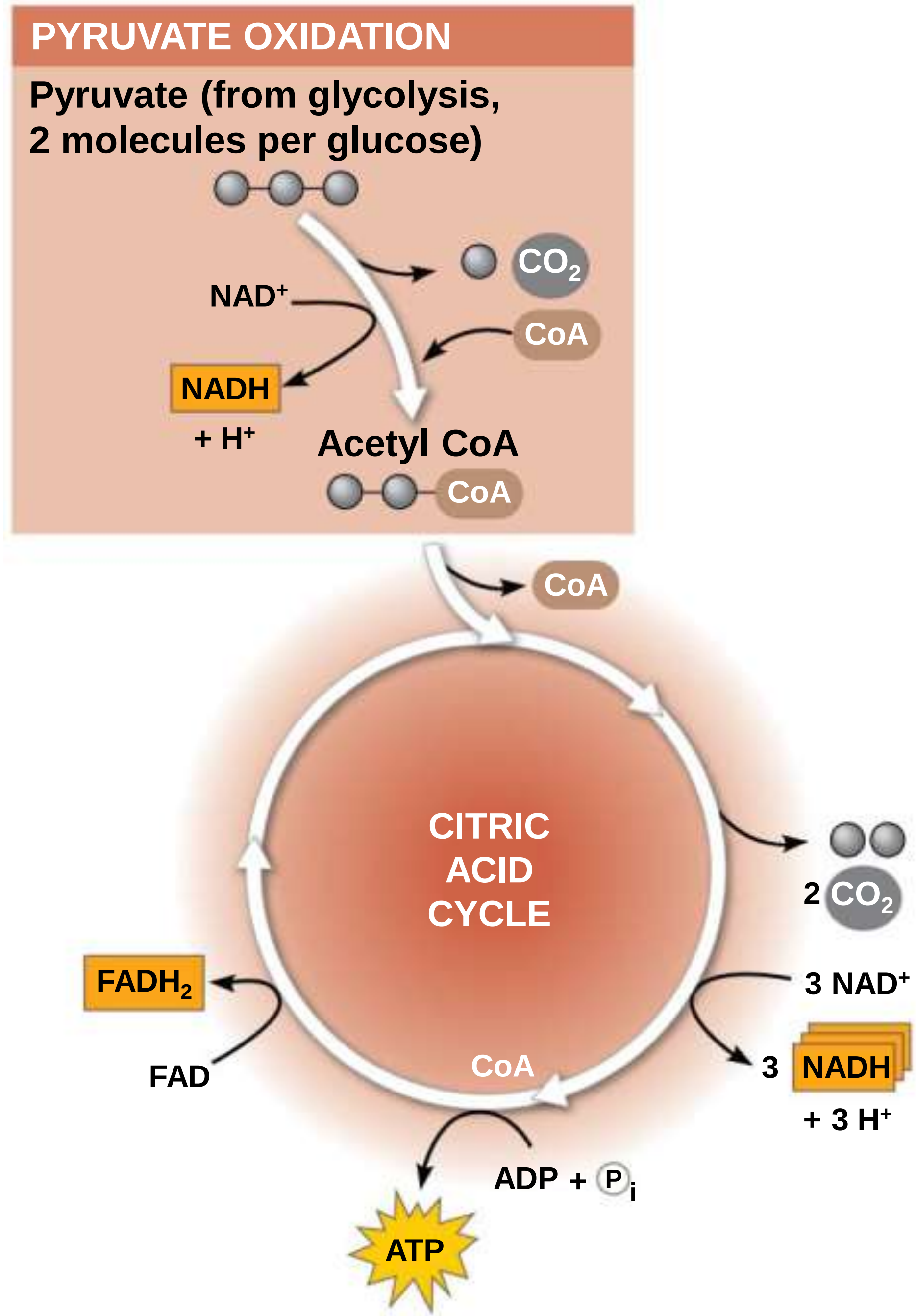

CYTOSOL
MITOCHONDRION
O⁻
C=O
C=O
CH3
Pyruvate
Transport protein
1
CO2
Coenzyme A
3
S-CoA
C=O
CH3
Acetyl CoA
2
NAD+
NADH
+ H+

The Citric Acid Cycle

a) The citric acid cycle, also called the Krebs cycle, completes the break down of pyruvate to CO_2

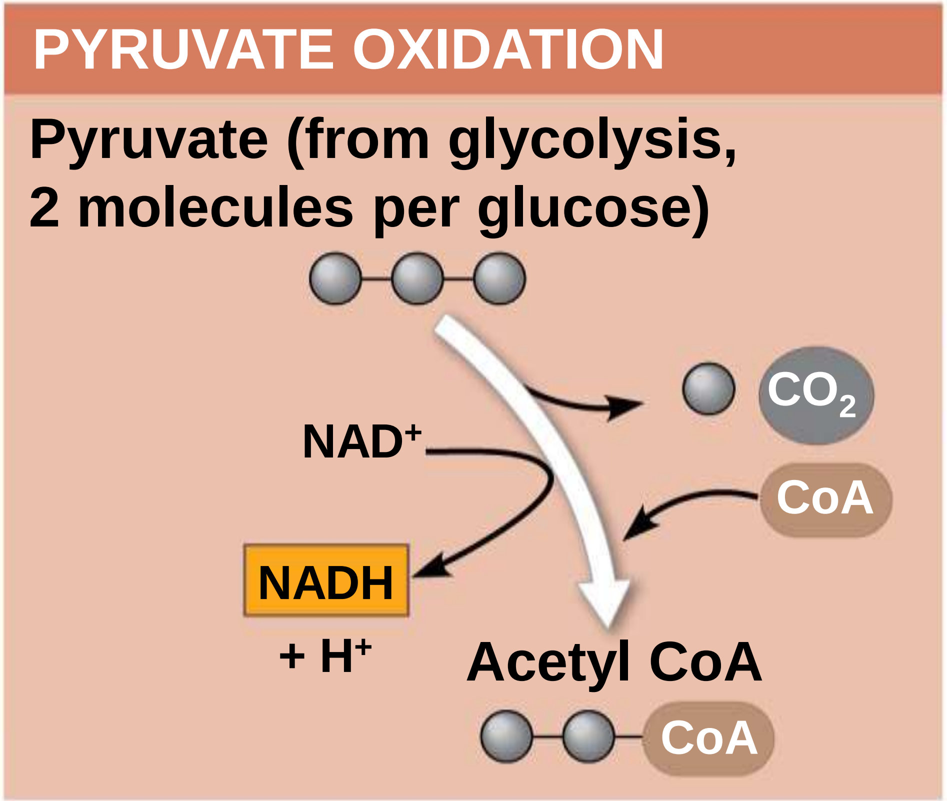
b) The cycle oxidizes organic fuel derived from pyruvate, generating 1 ATP, 3 NADH, and 1 $FADH_2$ per turn

Figure 9.11



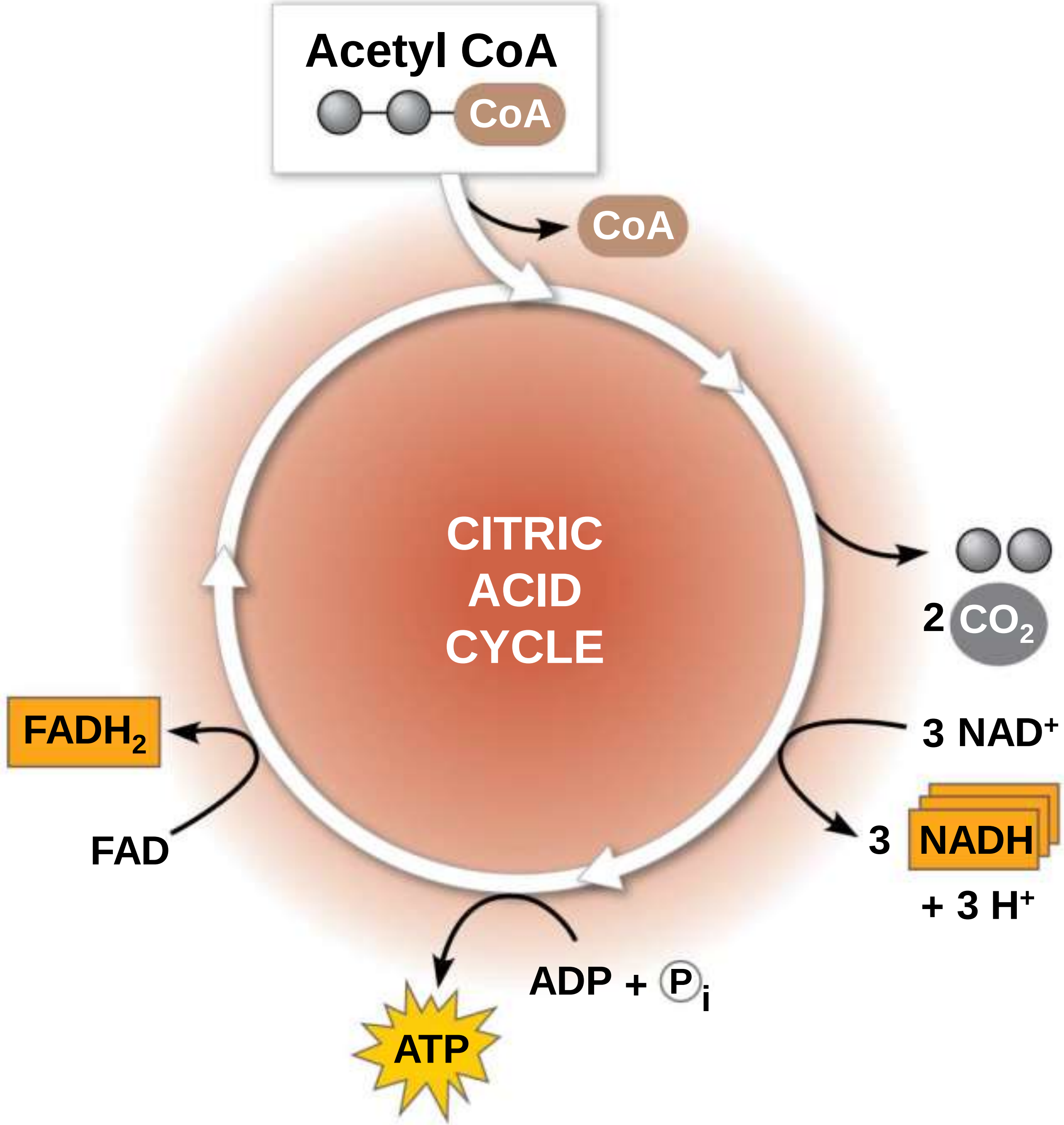
PYRUVATE OXIDATION
Pyruvate (from glycolysis, 2 molecules per glucose)
CO_2
NAD^+
CoA
NADH
$+ H^+$
Acetyl CoA
CoA
CoA
CITRIC ACID CYCLE
2 CO_2
3 NAD^+
3 NADH
$+ 3 H^+$
$FADH_2$
FAD
CoA
ADP + P i
ATP

Figure 9.11a



PYRUVATE OXIDATION
Pyruvate (from glycolysis, 2 molecules per glucose)
CO_2
NAD+
CoA
NADH
+ H+
Acetyl CoA
CoA

Figure 9.11b



Acetyl CoA
CoA
CoA
CITRIC
ACID
CYCLE
2 CO_2
3 NAD^+
3 NADH
+ 3 H^+
ADP + P i
ATP
FAD
$FADH_2$

a) The citric acid cycle has eight steps, each catalyzed by a specific enzyme

b) The acetyl group of acetyl CoA joins the cycle by combining with oxaloacetate, forming citrate

c) The next seven steps decompose the citrate back to oxaloacetate, making the process a cycle

d) The NADH and $FADH_2$ produced by the cycle relay electrons extracted from food to the electron transport chain

Figure 9.UN08

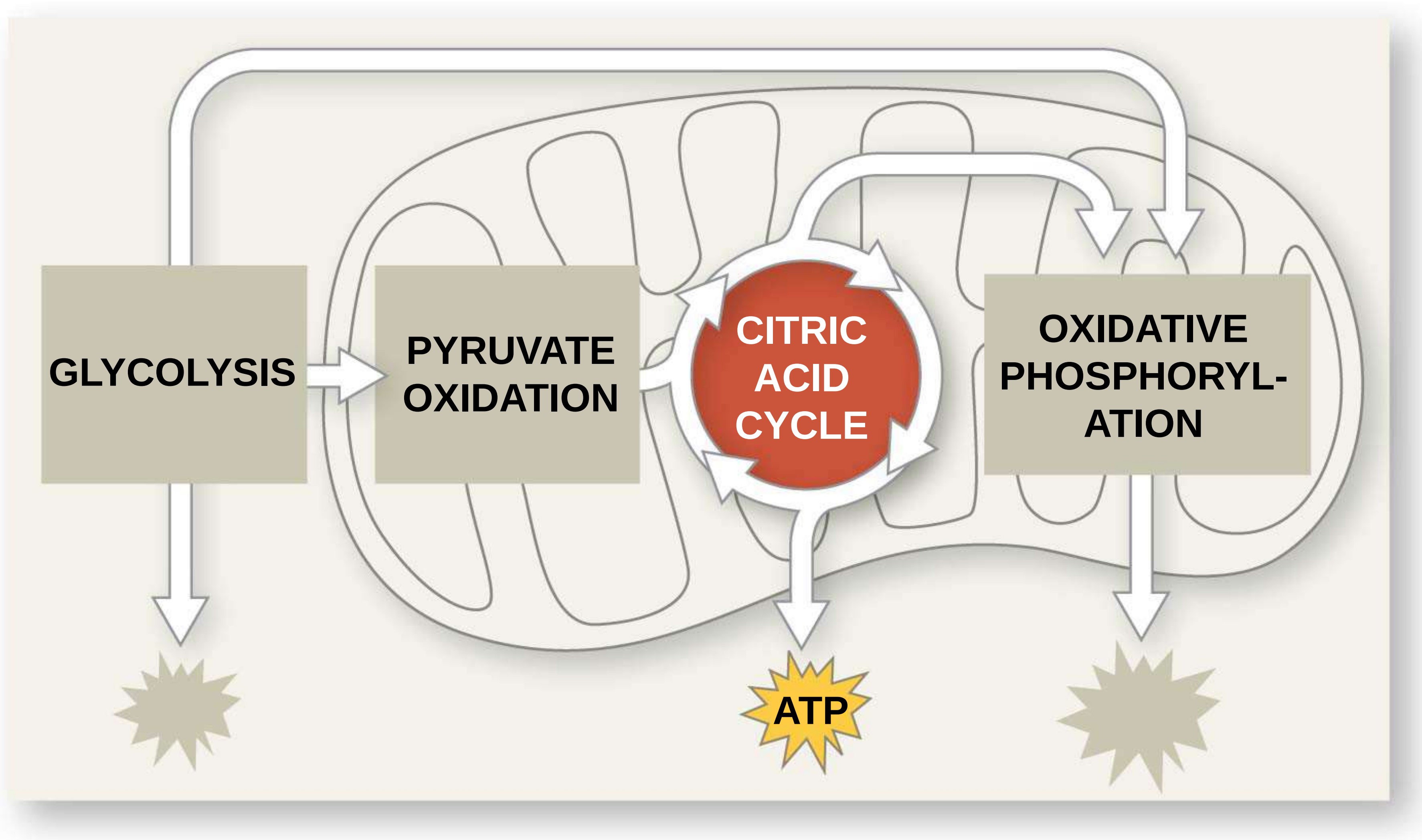
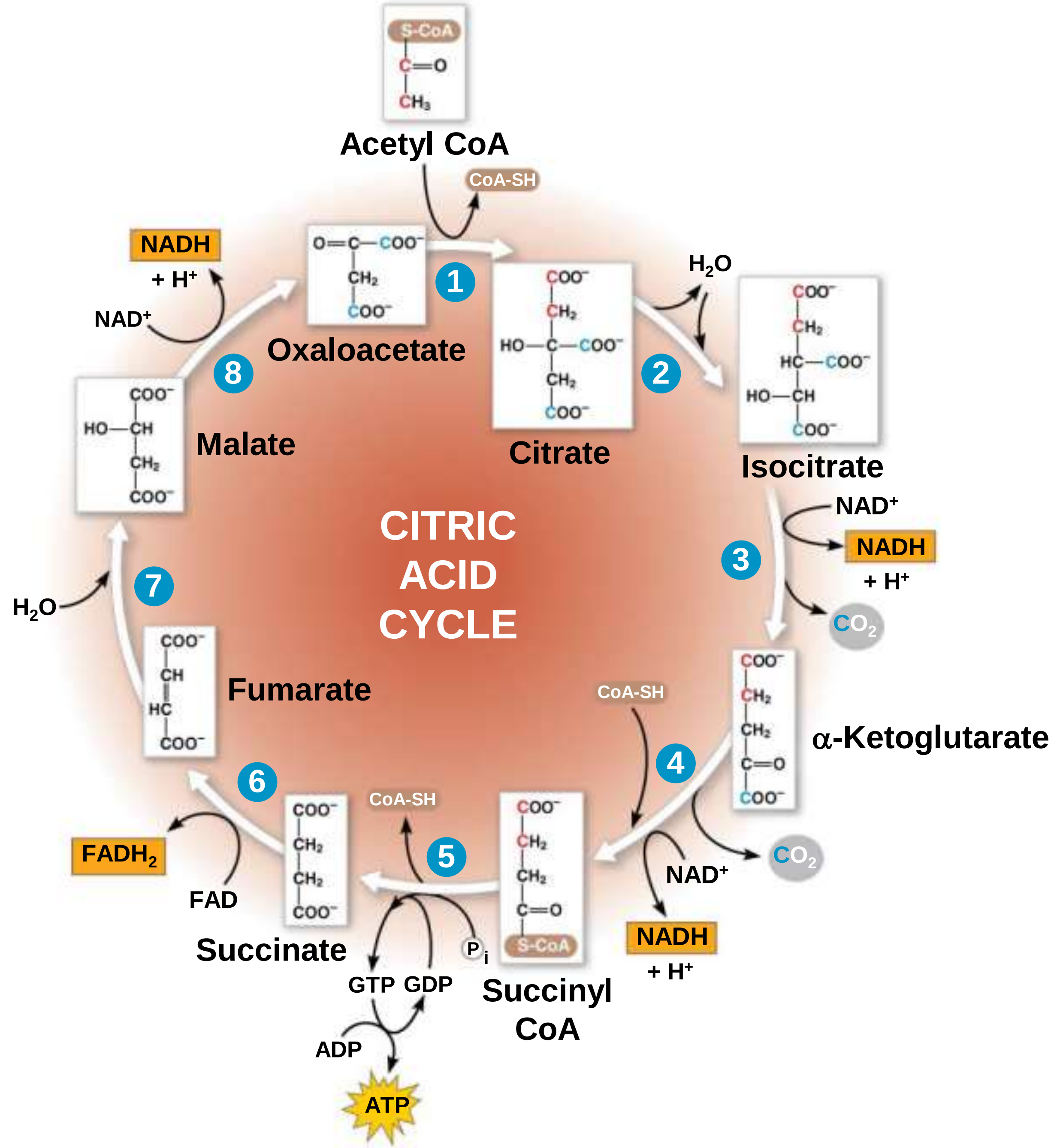


Figure 9.12-8



S-CoA
C=O
CH3
Acetyl CoA
CoA-SH
NADH
+ H+
NAD+
Oxaloacetate
Citrate
H2O
Isocitrate
1
2
3
4
5
6
7
8
Malate
CITRIC
ACID
CYCLE
NAD+
NADH
+ H+
CO2
H2O
Fumarate
CoA-SH
α-Ketoglutarate
CO2
NAD+
NADH
+ H+
CoA-SH
FADH2
FAD
Succinate
Pi
GTP GDP
Succinyl
CoA
ADP
ATP

Concept 9.4: During oxidative phosphorylation, chemiosmosis couples electron transport to ATP synthesis

a) Following glycolysis and the citric acid cycle, NADH and $FADH_2$ account for most of the energy extracted from food

b) These two electron carriers donate electrons to the electron transport chain, which powers ATP synthesis via oxidative phosphorylation

The Pathway of Electron Transport

a) The electron transport chain is in the inner membrane (cristae) of the mitochondrion

b) Most of the chain's components are proteins, which exist in multiprotein complexes

c) The carriers alternate reduced and oxidized states as they accept and donate electrons

d) Electrons drop in free energy as they go down the chain and are finally passed to O_2, forming H_2O

Figure 9.UN09

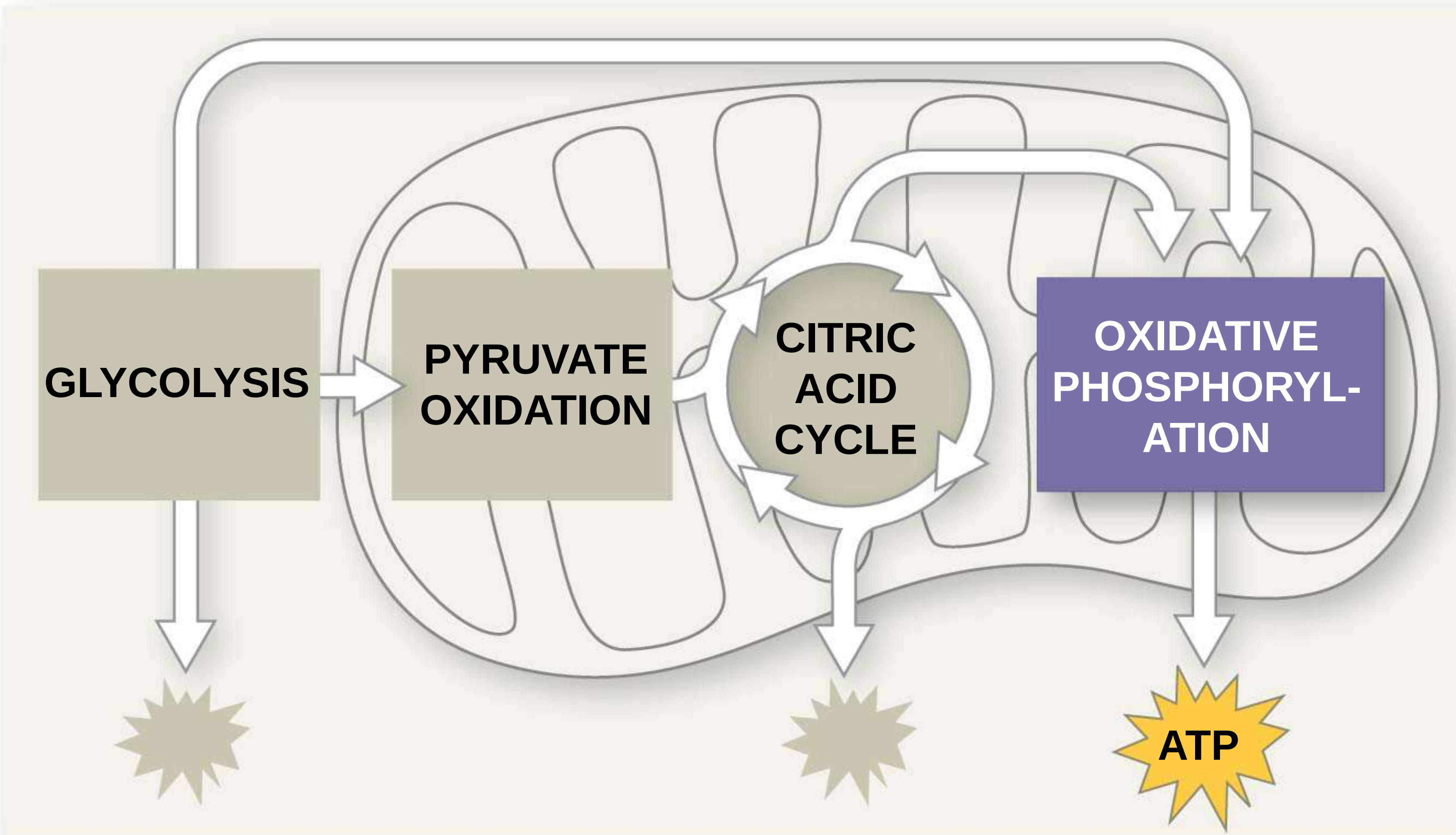
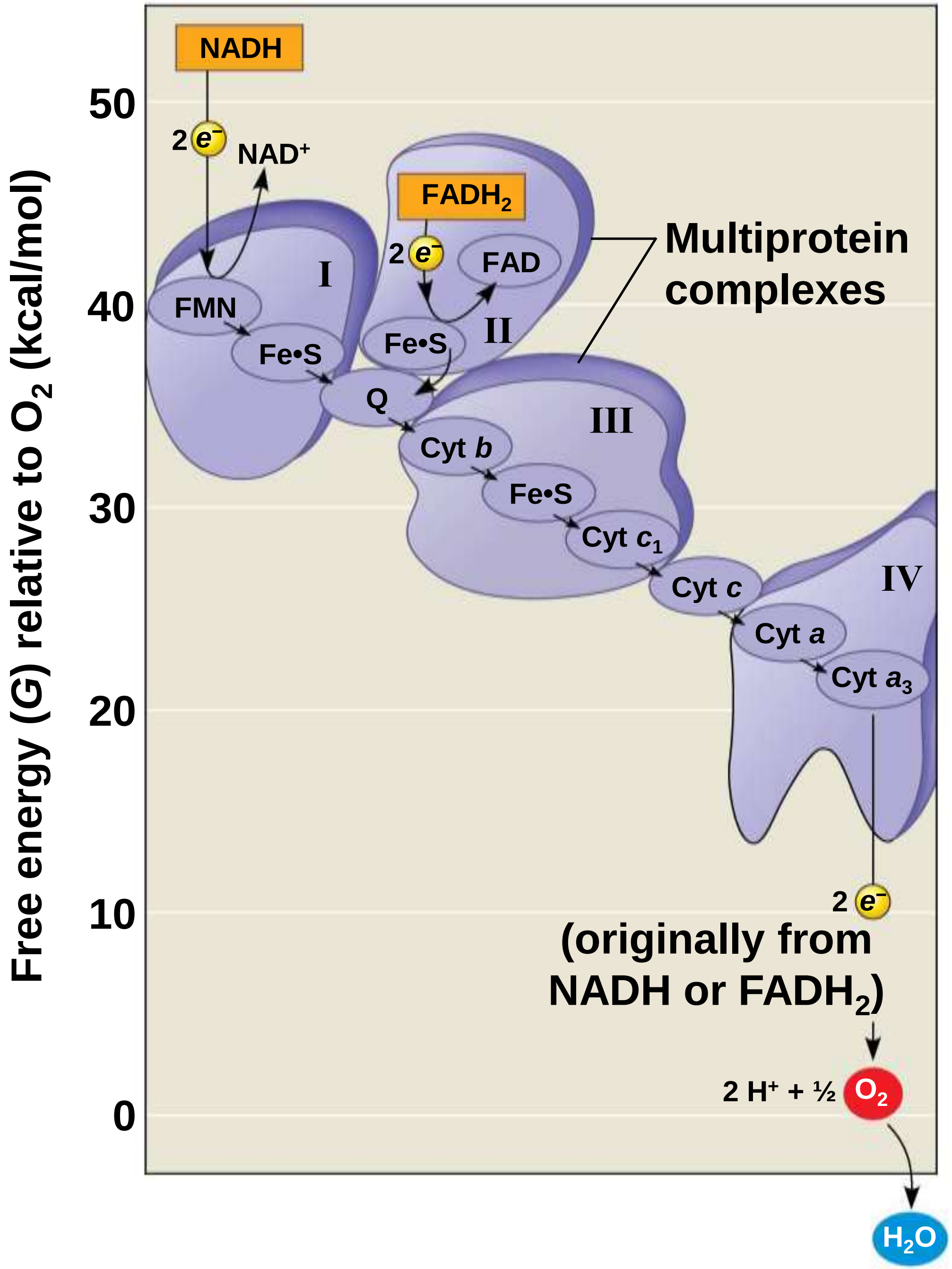
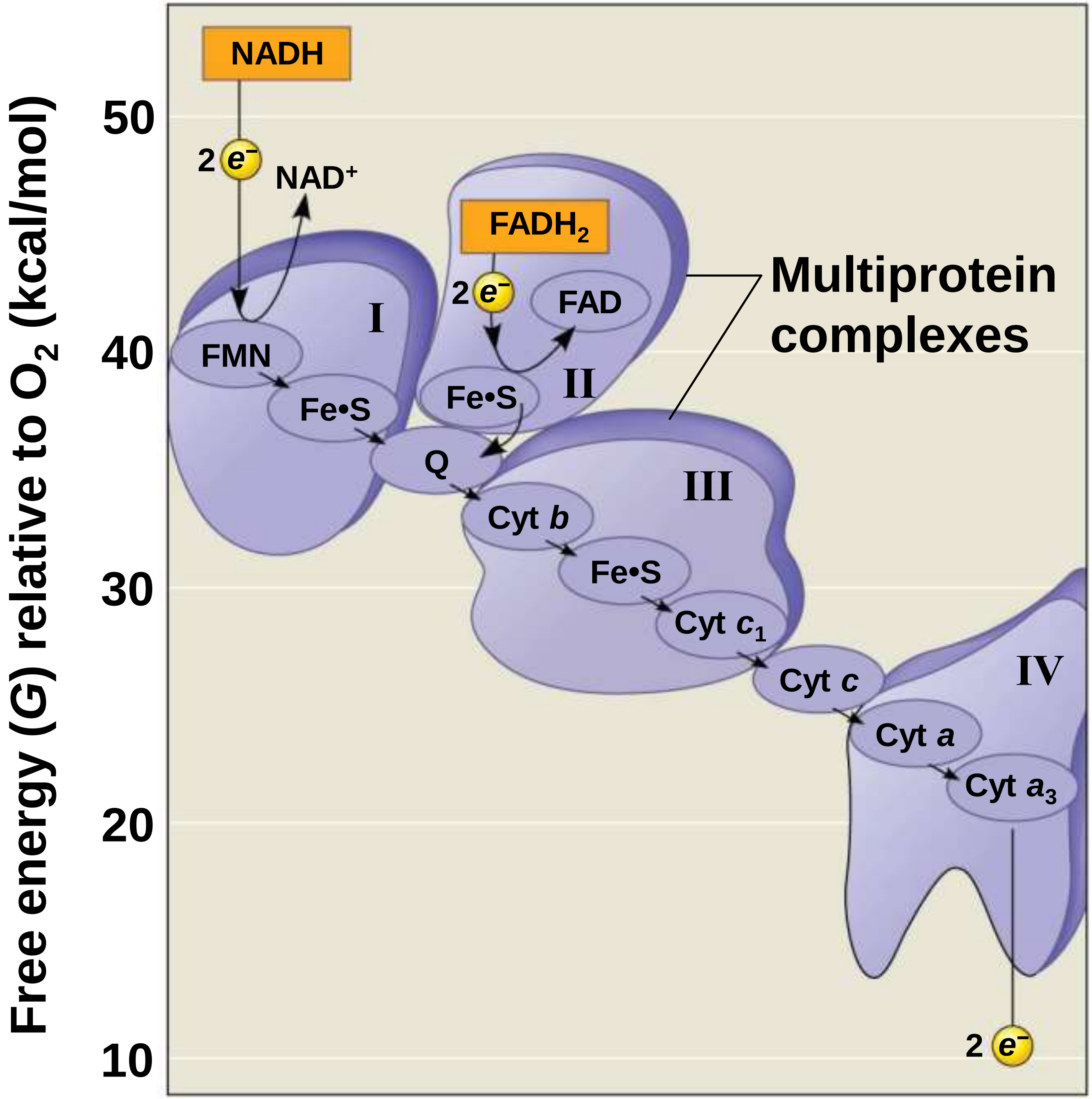


Figure 9.13



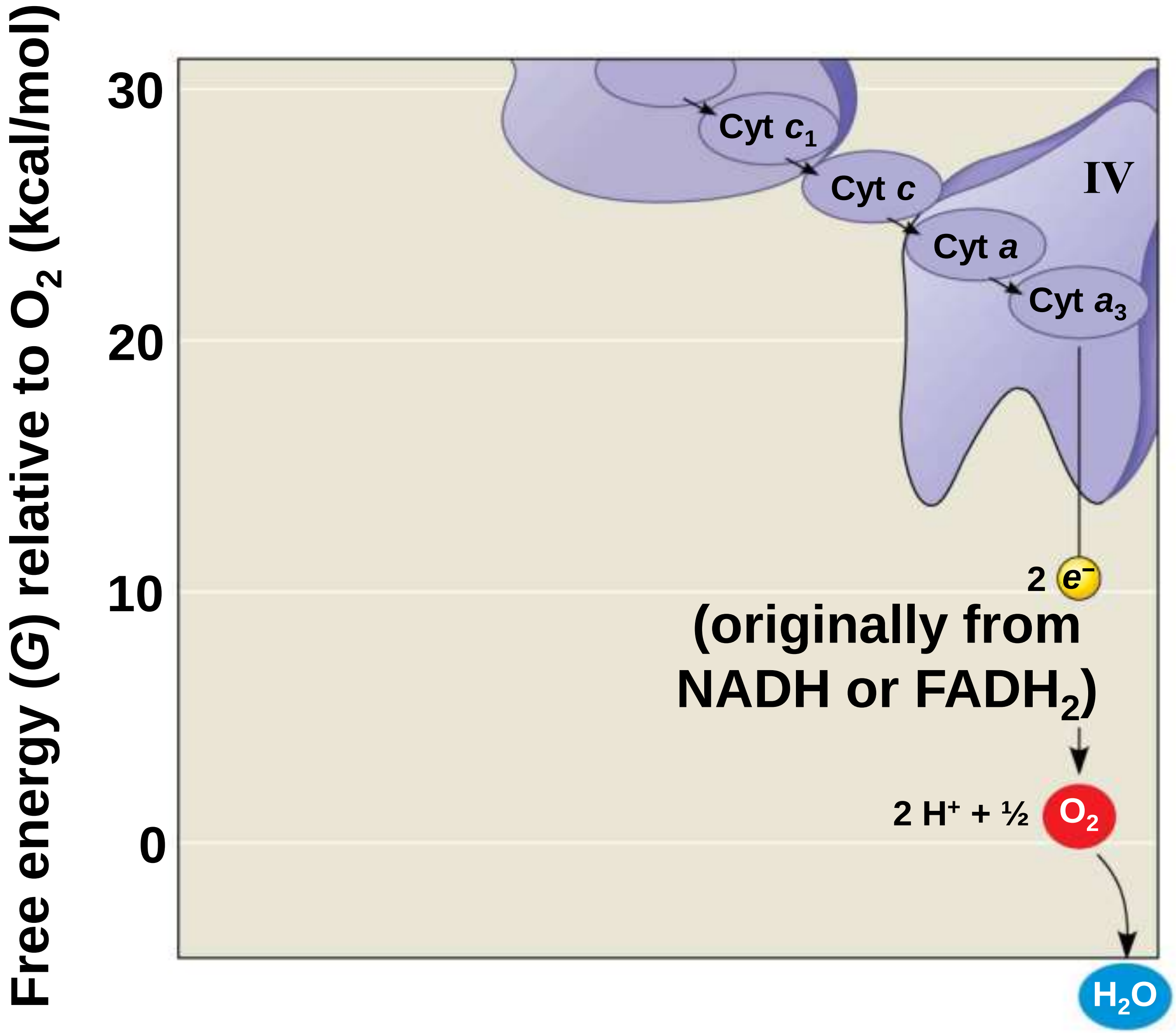
NADH
50
2 e^-
NAD^+
$FADH_2$
Multiprotein complexes
2 e^-
FAD
40
FMN
I
Fe•S
II
Fe•S
Q
III
Cyt *b*
30
Fe•S
Cyt c_1
Cyt *c*
IV
Cyt *a*
Cyt a_3
20
10
2 e^-
(originally from NADH or $FADH_2$)
2 H^+ + ½ O_2
0
H_2O
Free energy (*G*) relative to O_2 (kcal/mol)

Figure 9.13a



NADH
50
2 e⁻
NAD⁺
FADH₂
Multiprotein complexes
2 e⁻
FAD
I
40
FMN
Fe•S
II
Fe•S
Q
III
Cyt b
Fe•S
30
Cyt c₁
Cyt c
IV
Cyt a
Cyt a₃
20
2 e⁻
10
Free energy (G) relative to O₂ (kcal/mol)

Figure 9.13b



Free energy (G) relative to O2 (kcal/mol)
30
20
10
0
Cyt c1
Cyt c
IV
Cyt a
Cyt a3
2 e−
(originally from NADH or FADH2)
2 H+ + ½ O2
H2O

a) Electrons are transferred from NADH or $FADH_2$ to the electron transport chain

b) Electrons are passed through a number of proteins including **cytochromes** (each with an iron atom) to O_2

c) The electron transport chain generates no ATP directly

d) It breaks the large free-energy drop from food to O_2 into smaller steps that release energy in manageable amounts

Chemiosmosis: The Energy-Coupling Mechanism

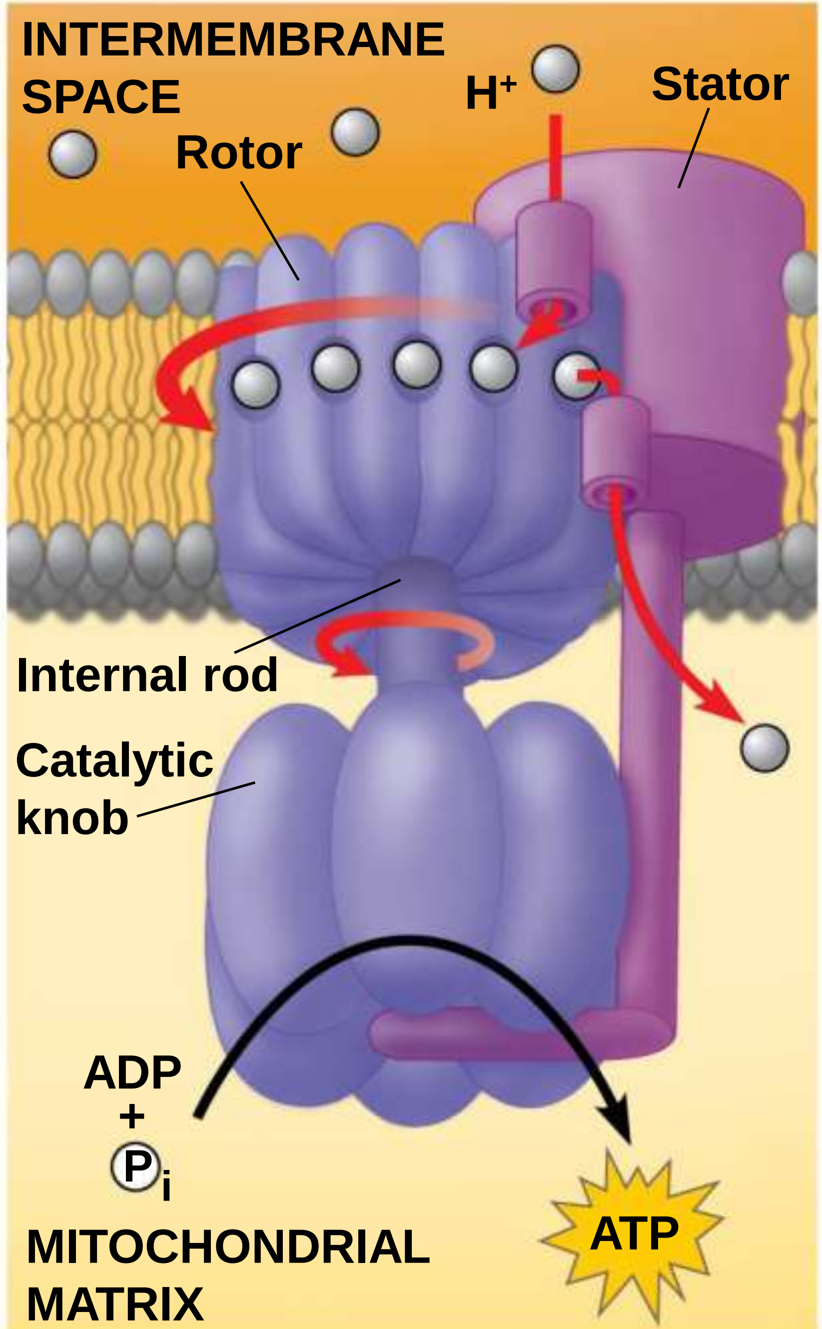
a) Electron transfer in the electron transport chain causes proteins to pump H^+ from the mitochondrial matrix to the intermembrane space

b) H+ then moves back across the membrane, passing through the protein complex, **ATP synthase**

c) ATP synthase uses the exergonic flow of H^+ to drive phosphorylation of ATP

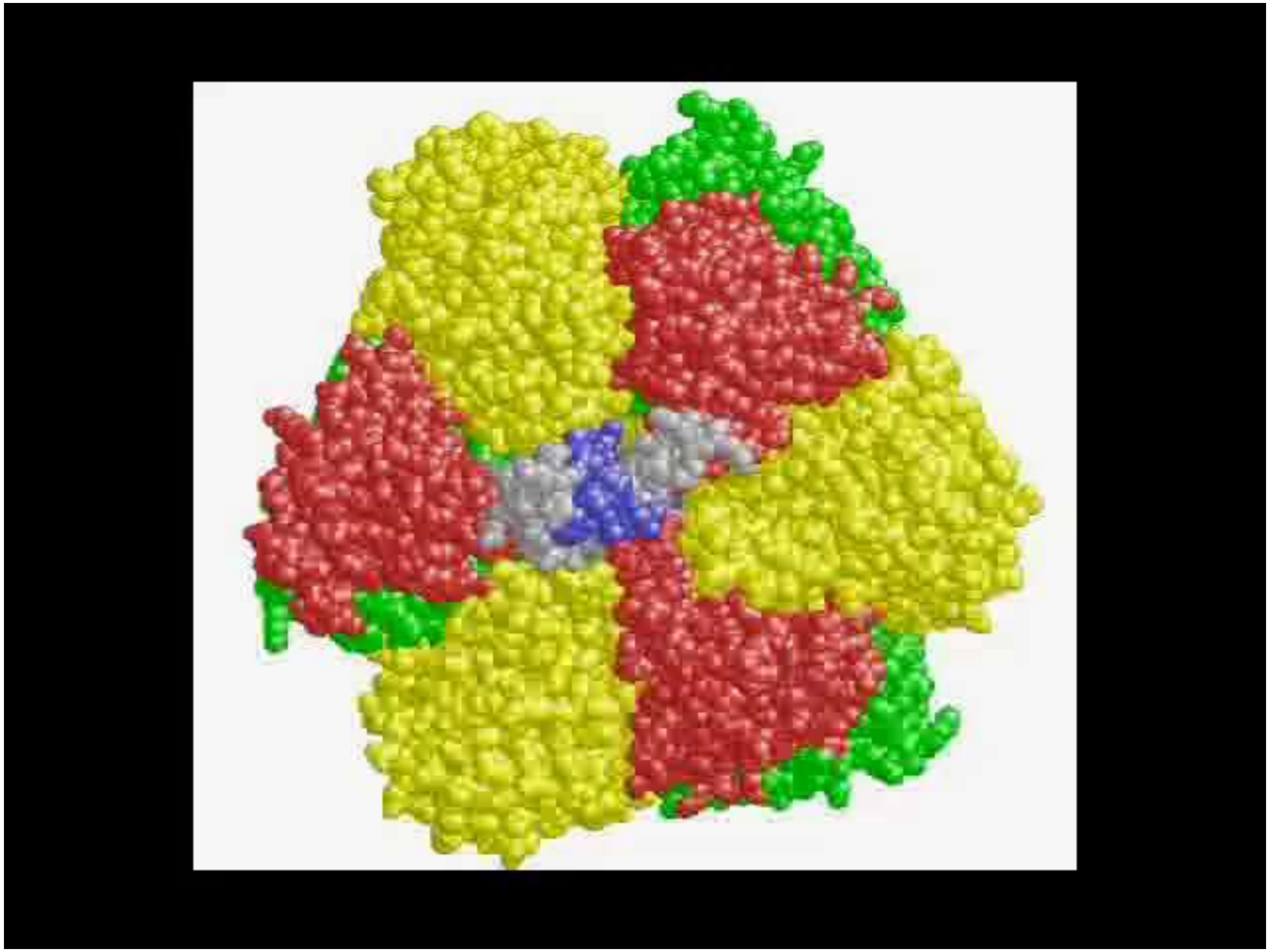
d) This is an example of **chemiosmosis**, the use of energy in a H^+ gradient to drive cellular work

Figure 9.14



INTERMEMBRANE SPACE
H+
Stator
Rotor
Internal rod
Catalytic knob
ADP
+
P i
ATP
MITOCHONDRIAL MATRIX

Video: ATP Synthase 3-D Structure, Top View



Video: ATP Synthase 3-D Structure, Side View

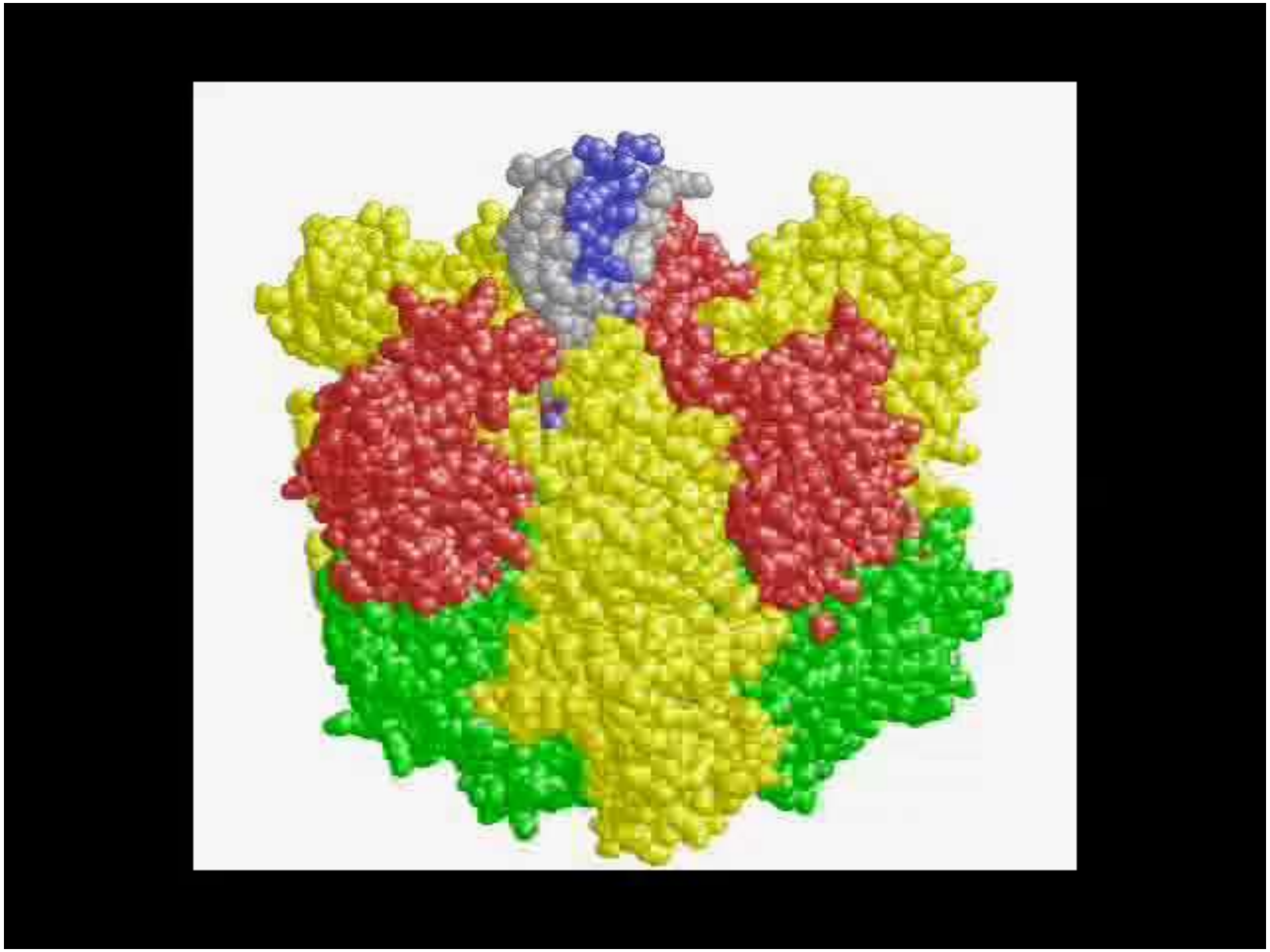
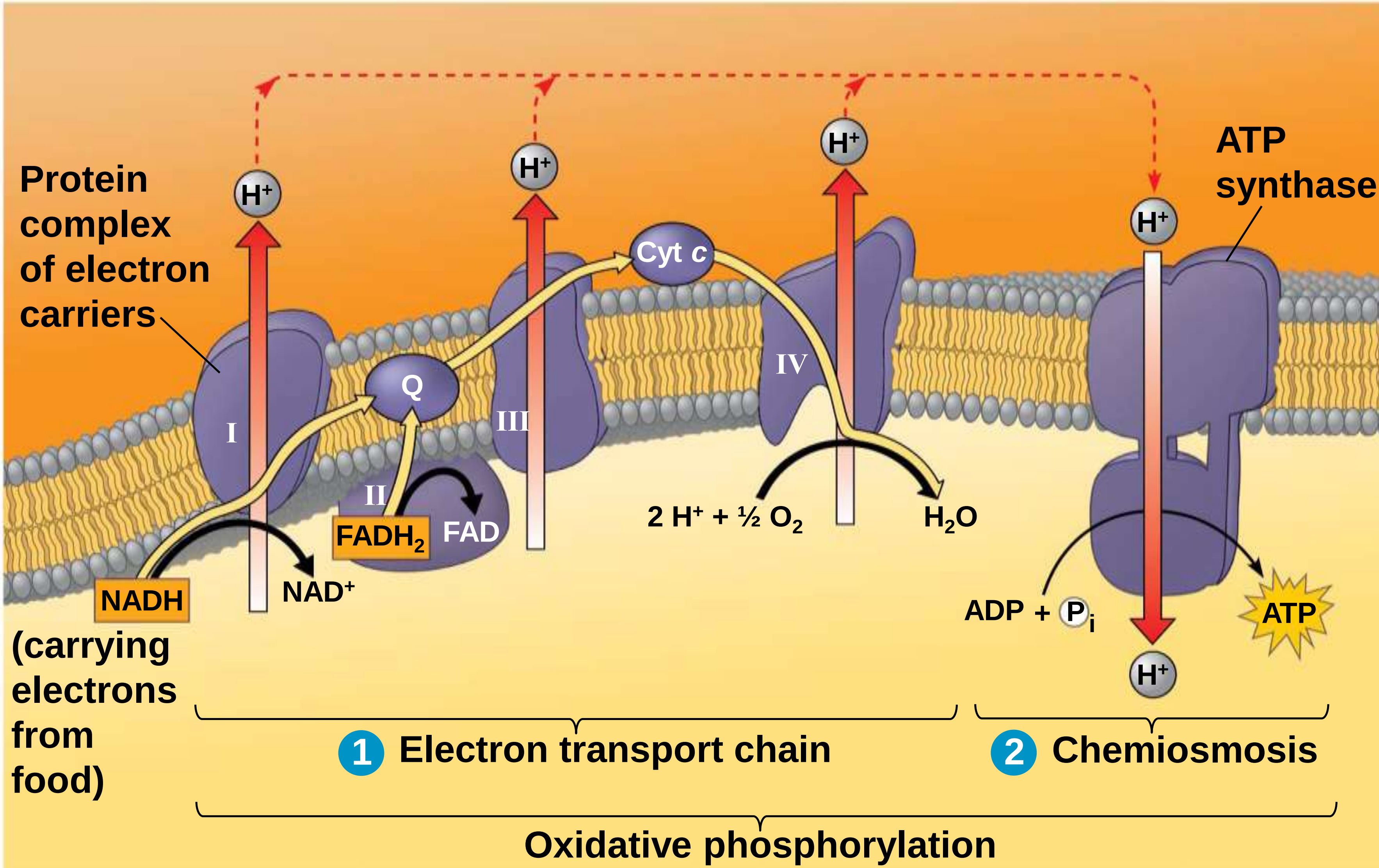
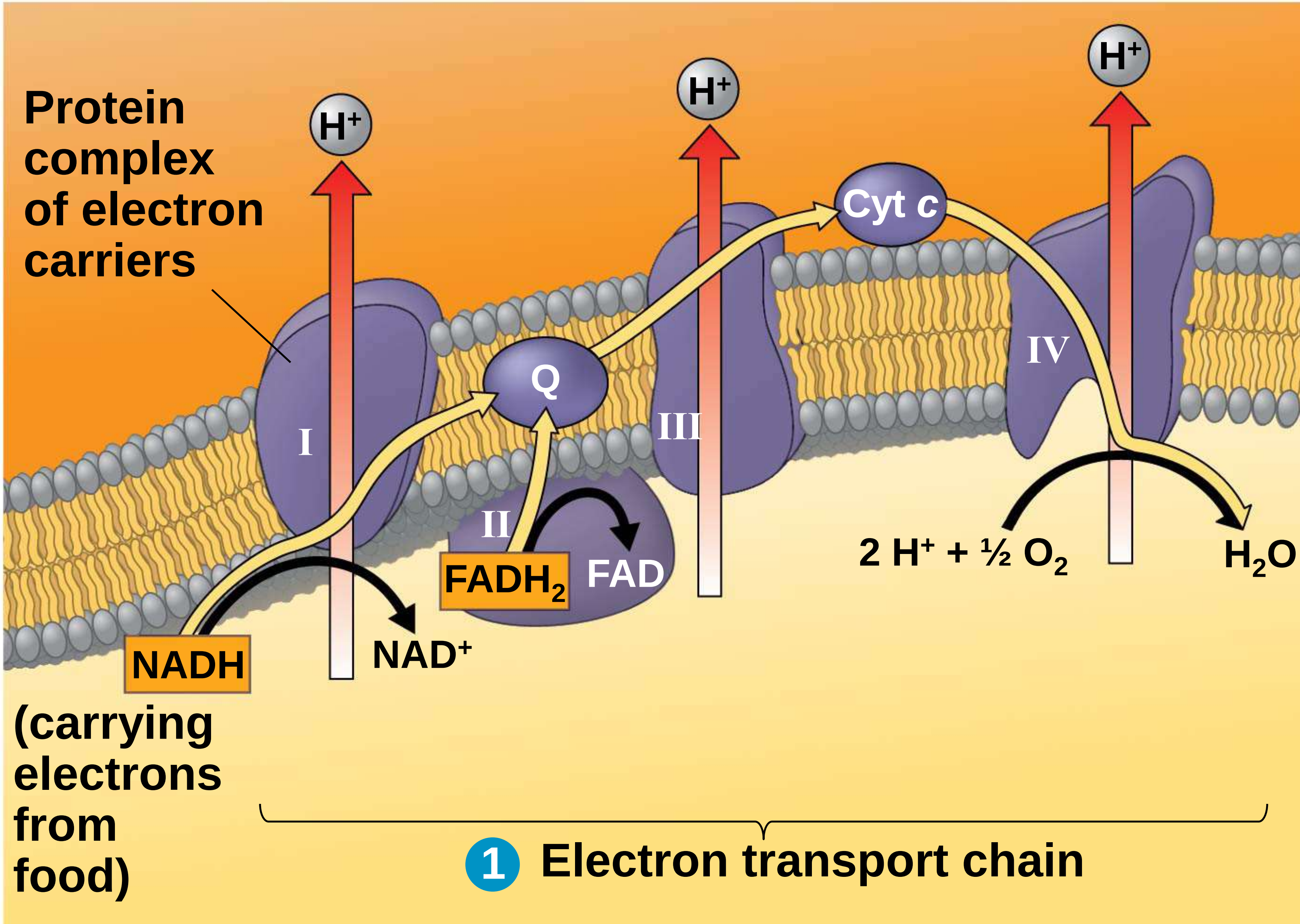


Figure 9.15



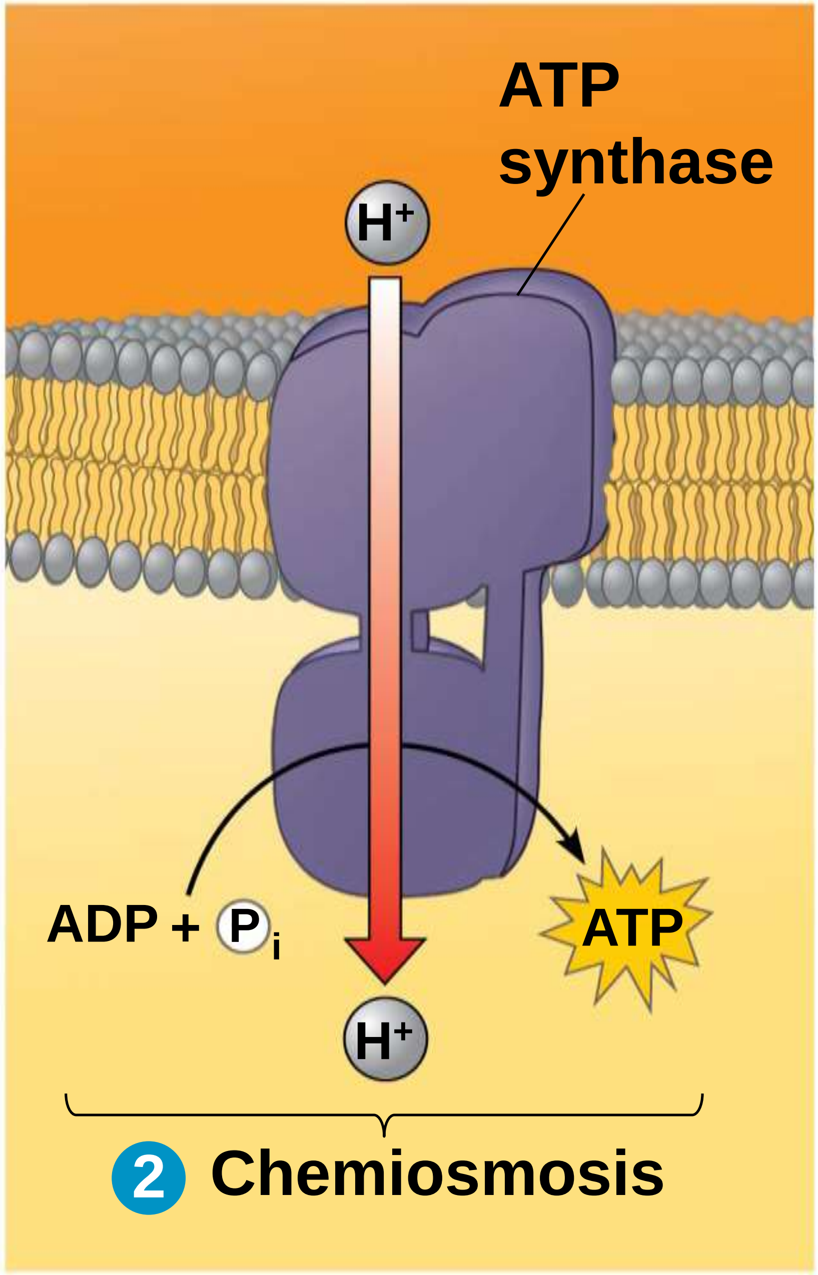
Protein complex of electron carriers
ATP synthase
H+
H+
H+
H+
H+
Cyt c
Q
I
II
III
IV
NADH
NAD+
FADH2
FAD
2 H+ + ½ O2
H2O
ADP + P i
ATP
(carrying electrons from food)
1 Electron transport chain
2 Chemiosmosis
Oxidative phosphorylation

Figure 9.15a



Protein complex of electron carriers
H+
H+
H+
Cyt c
Q
I
II
III
IV
FADH2
FAD
NADH
NAD+
2 H+ + ½ O2
H2O
(carrying electrons from food)
1 Electron transport chain

Figure 9.15b



a) The energy stored in a H^+ gradient across a membrane couples the redox reactions of the electron transport chain to ATP synthesis

b) The H+ gradient is referred to as a **proton-motive force**, emphasizing its capacity to do work

An Accounting of ATP Production by Cellular Respiration

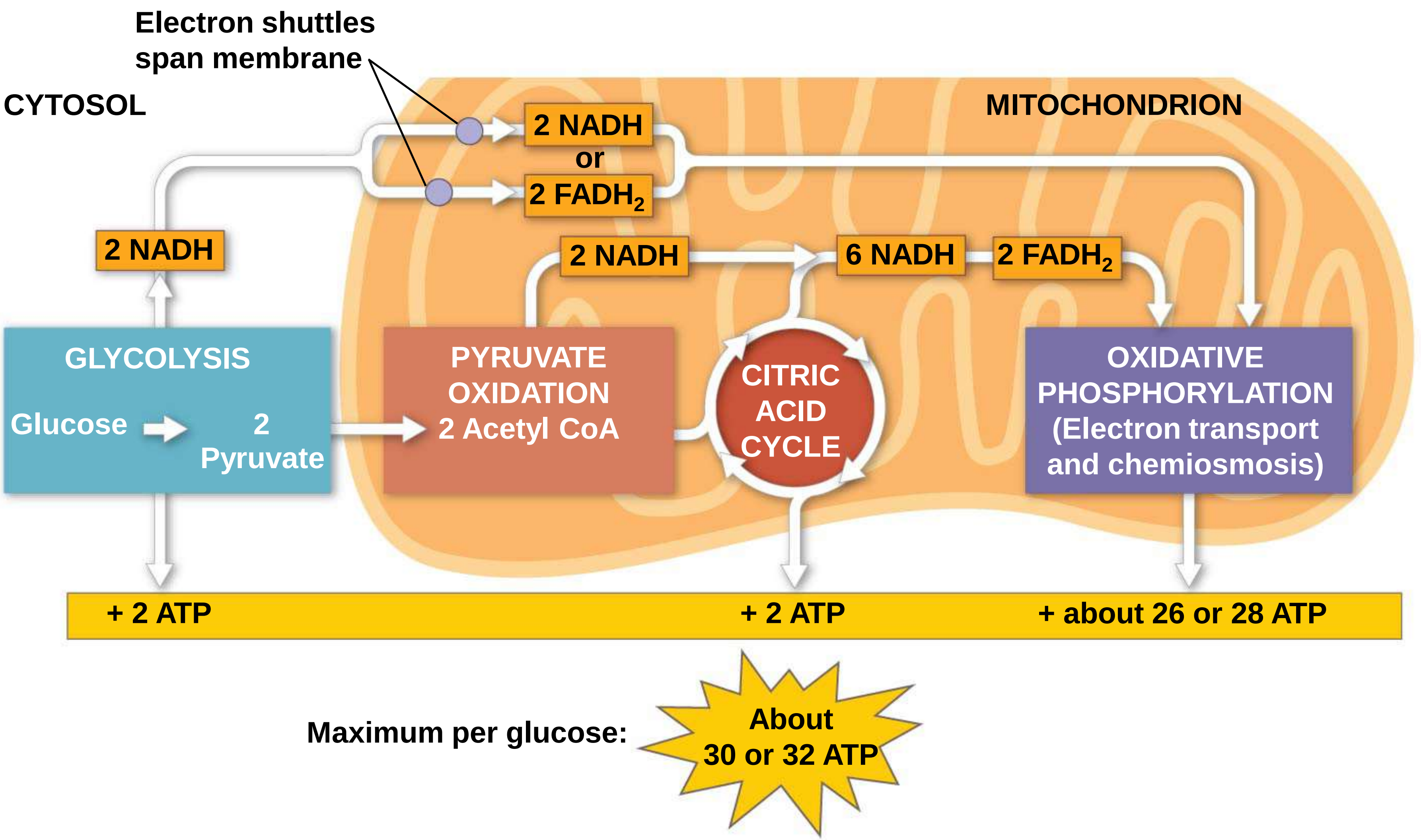
a) During cellular respiration, most energy flows in this sequence:

   glucose → NADH → electron transport chain → proton-motive force → ATP

a) About 34% of the energy in a glucose molecule is transferred to ATP during cellular respiration, making about 32 ATP

b) There are several reasons why the number of ATP is not known exactly

Figure 9.16



Electron shuttles span membrane
CYTOSOL
MITOCHONDRION
2 NADH
or
2 $FADH_2$
2 NADH
2 NADH
6 NADH
2 $FADH_2$
GLYCOLYSIS
Glucose
2 Pyruvate
PYRUVATE OXIDATION
2 Acetyl CoA
CITRIC ACID CYCLE
OXIDATIVE PHOSPHORYLATION (Electron transport and chemiosmosis)
+ 2 ATP
+ 2 ATP
+ about 26 or 28 ATP
Maximum per glucose:
About 30 or 32 ATP

Figure 9.16a

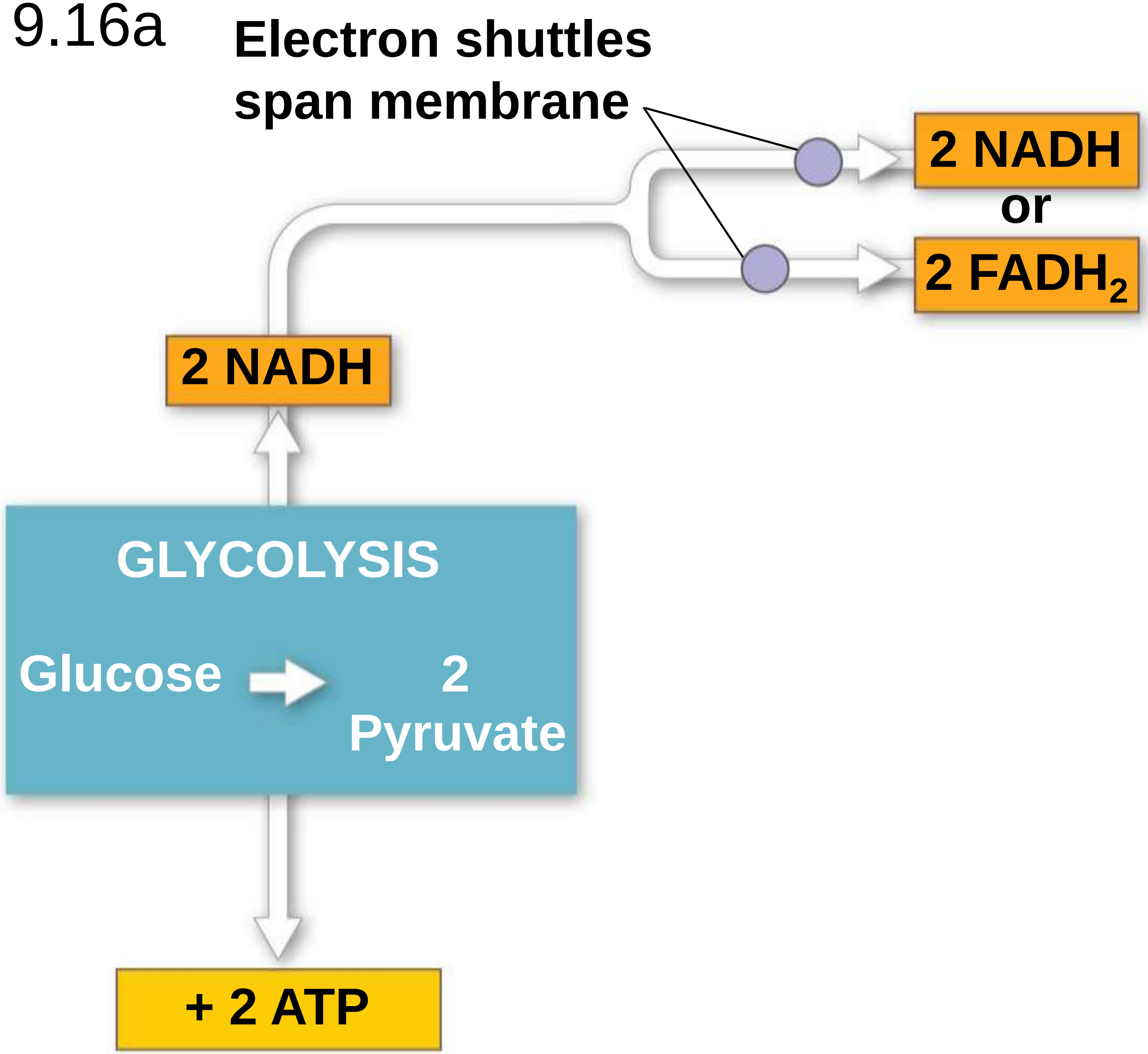

Electron shuttles span membrane
2 NADH
or
2 FADH2
2 NADH
GLYCOLYSIS
Glucose
2 Pyruvate
+ 2 ATP

Figure 9.16b

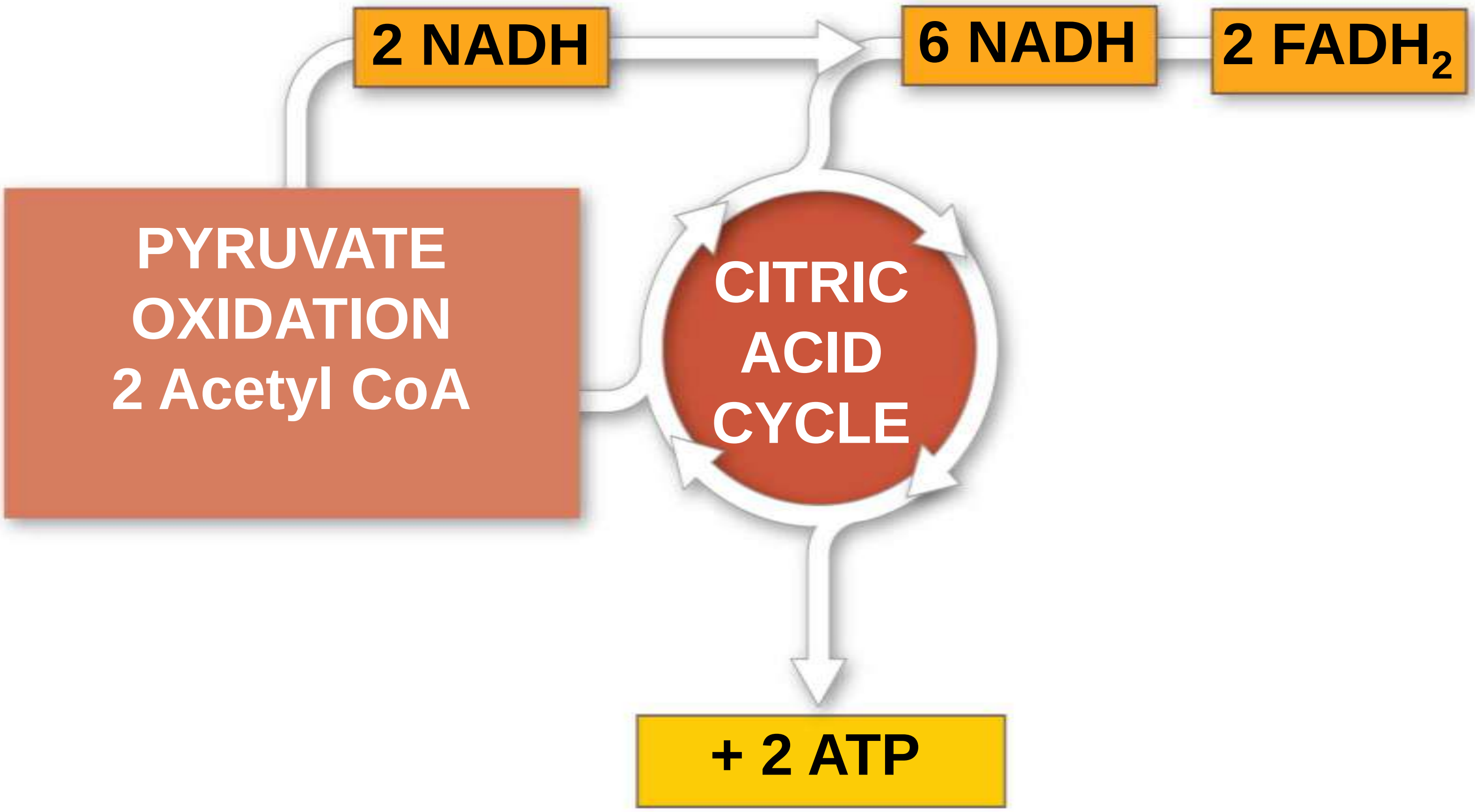

2 NADH
6 NADH
2 $FADH_2$
PYRUVATE OXIDATION
2 Acetyl CoA
CITRIC ACID CYCLE
+ 2 ATP

Figure 9.16c

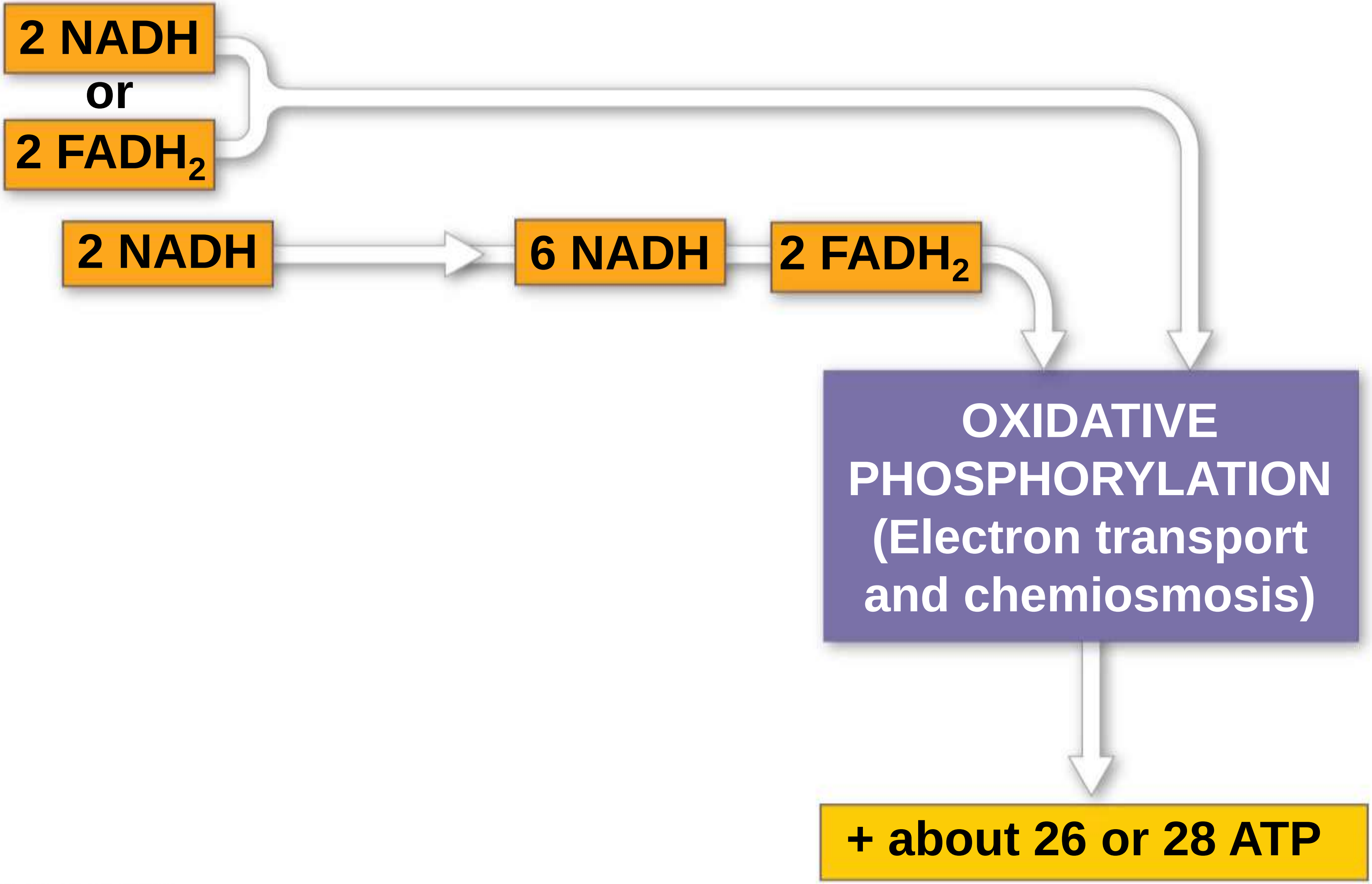

2 NADH
or
2 $FADH_2$
2 NADH
6 NADH
2 $FADH_2$
OXIDATIVE PHOSPHORYLATION (Electron transport and chemiosmosis)
+ about 26 or 28 ATP

Figure 9.16d

Maximum per glucose:

About
30 or 32 ATP

Concept 9.5: Fermentation and anaerobic respiration enable cells to produce ATP without the use of oxygen

a) Most cellular respiration requires O_2 to produce ATP

b) Without O_2, the electron transport chain will cease to operate

c) In that case, glycolysis couples with anaerobic respiration or fermentation to produce ATP

a) Anaerobic respiration uses an electron transport chain with a final electron acceptor other than O_2, for example sulfate

b) Fermentation uses substrate-level phosphorylation instead of an electron transport chain to generate ATP

Types of Fermentation

a) Fermentation consists of glycolysis plus reactions that regenerate NAD^+, which can be reused by glycolysis

b) Two common types are alcohol fermentation and lactic acid fermentation

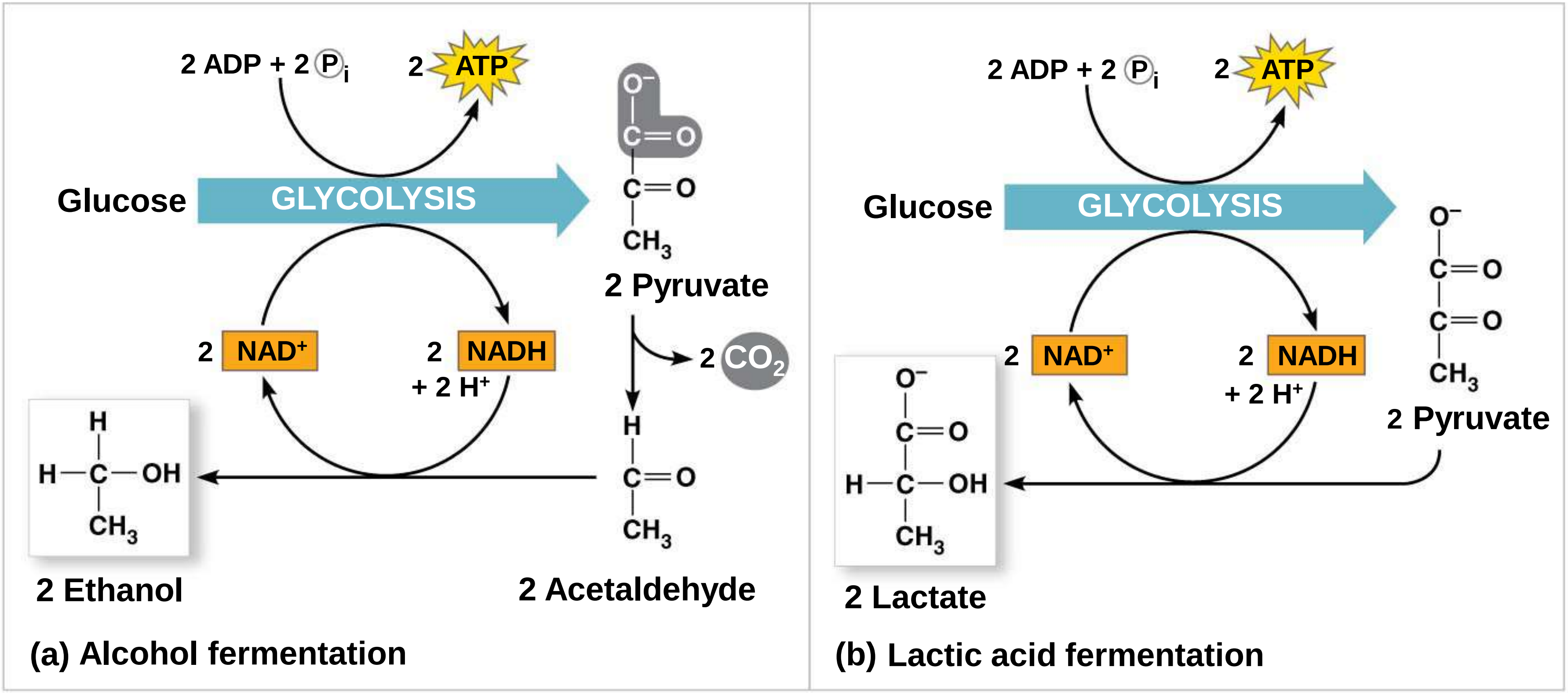
a) In **alcohol fermentation**, pyruvate is converted to ethanol in two steps

    a) The first step releases CO_2

    b) The second step produces ethanol

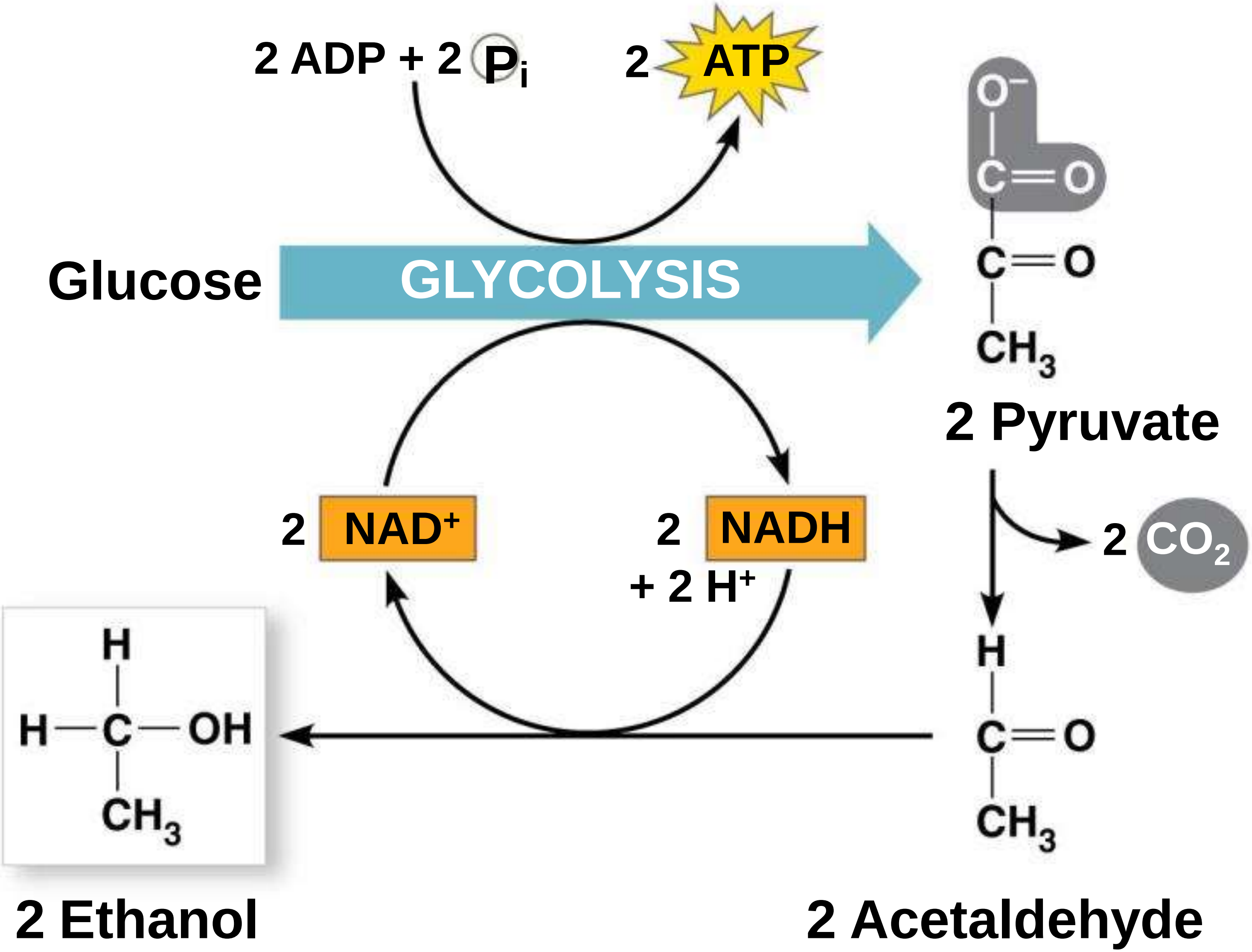
b) Alcohol fermentation by yeast is used in brewing, winemaking, and baking

Figure 9.17



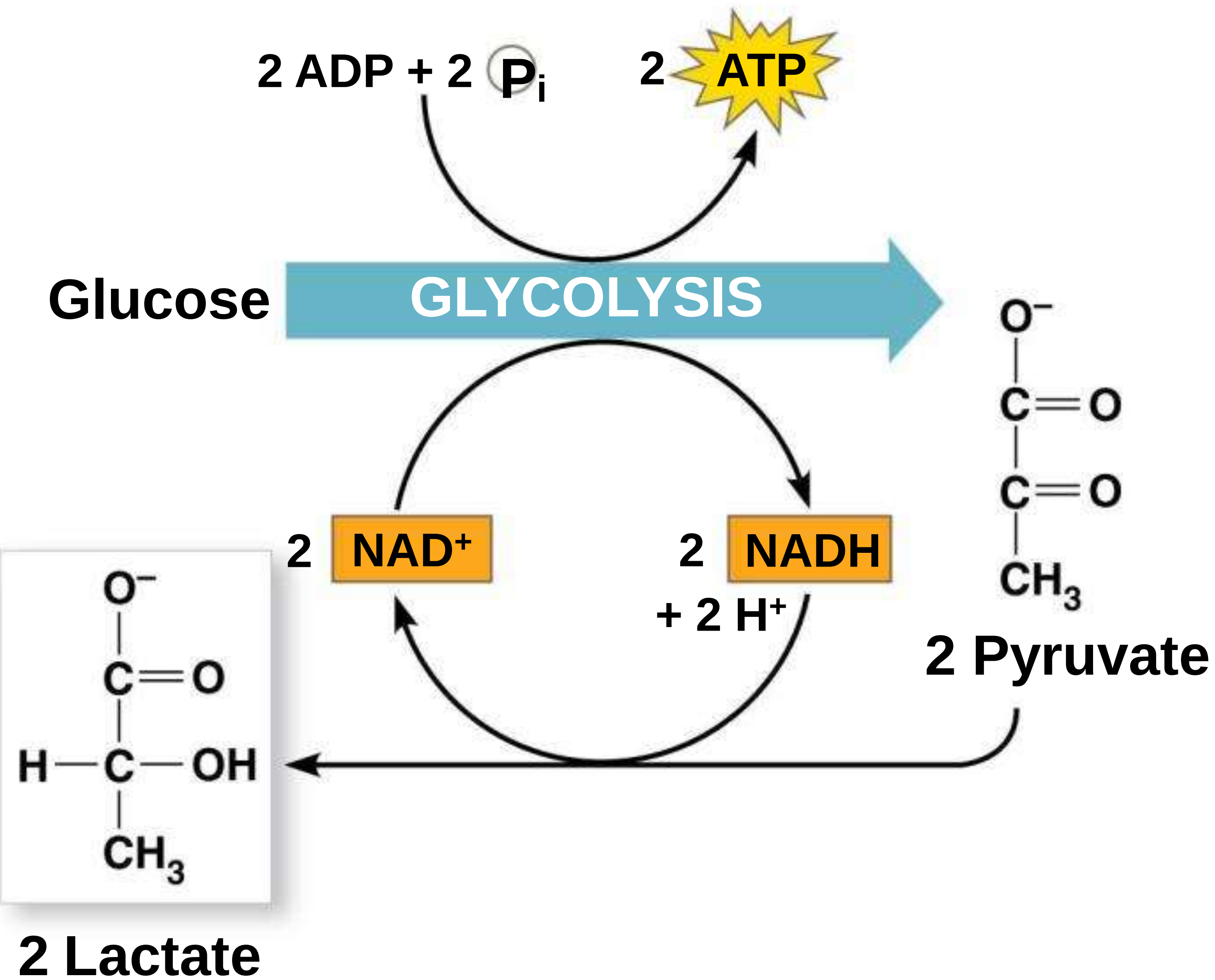
2 ADP + 2 P i
2 ATP
Glucose
GLYCOLYSIS
O−
C=O
C=O
CH3
2 Pyruvate
2 NAD+
2 NADH
+ 2 H+
2 CO2
H
C=O
CH3
H
H—C—OH
CH3
2 Ethanol
2 Acetaldehyde
(a) Alcohol fermentation
2 ADP + 2 P i
2 ATP
Glucose
GLYCOLYSIS
O−
C=O
C=O
CH3
2 Pyruvate
2 NAD+
2 NADH
+ 2 H+
O−
C=O
H—C—OH
CH3
2 Lactate
(b) Lactic acid fermentation

Figu



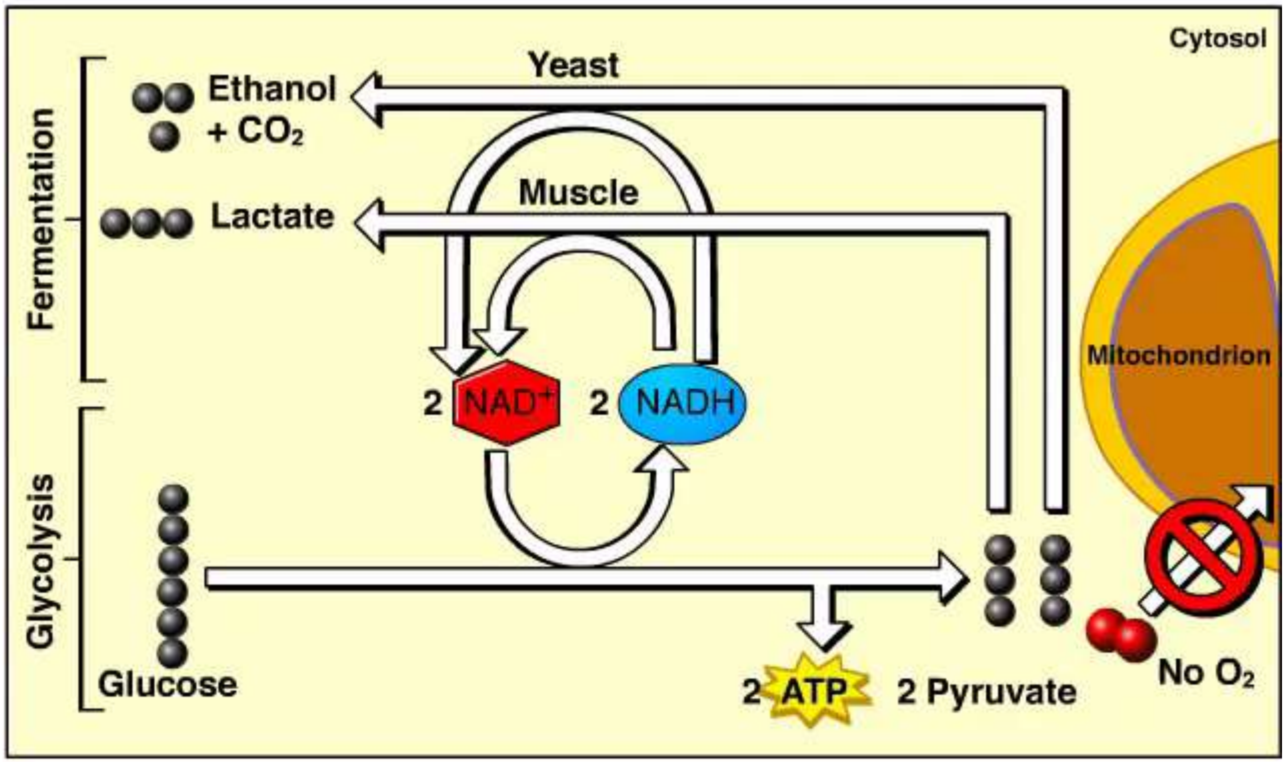
(a) Alcohol fermentation

Figure



(b) Lactic acid fermentation

Animation: Fermentation Overview



a) In **lactic acid fermentation**, pyruvate is reduced by NADH, forming lactate as an end product, with no release of CO_2

b) Lactic acid fermentation by some fungi and bacteria is used to make cheese and yogurt

c) Human muscle cells use lactic acid fermentation to generate ATP when O_2 is scarce

Comparing Fermentation with Anaerobic and Aerobic Respiration

a) All use glycolysis (net ATP = 2) to oxidize glucose and harvest chemical energy of food

b) In all three, NAD^+ is the oxidizing agent that accepts electrons during glycolysis

a) The processes have different mechanisms for oxidizing NADH:

    a) In fermentation, an organic molecule (such as pyruvate or acetaldehyde) acts as a final electron acceptor

    b) In cellular respiration electrons are transferred to the electron transport chain

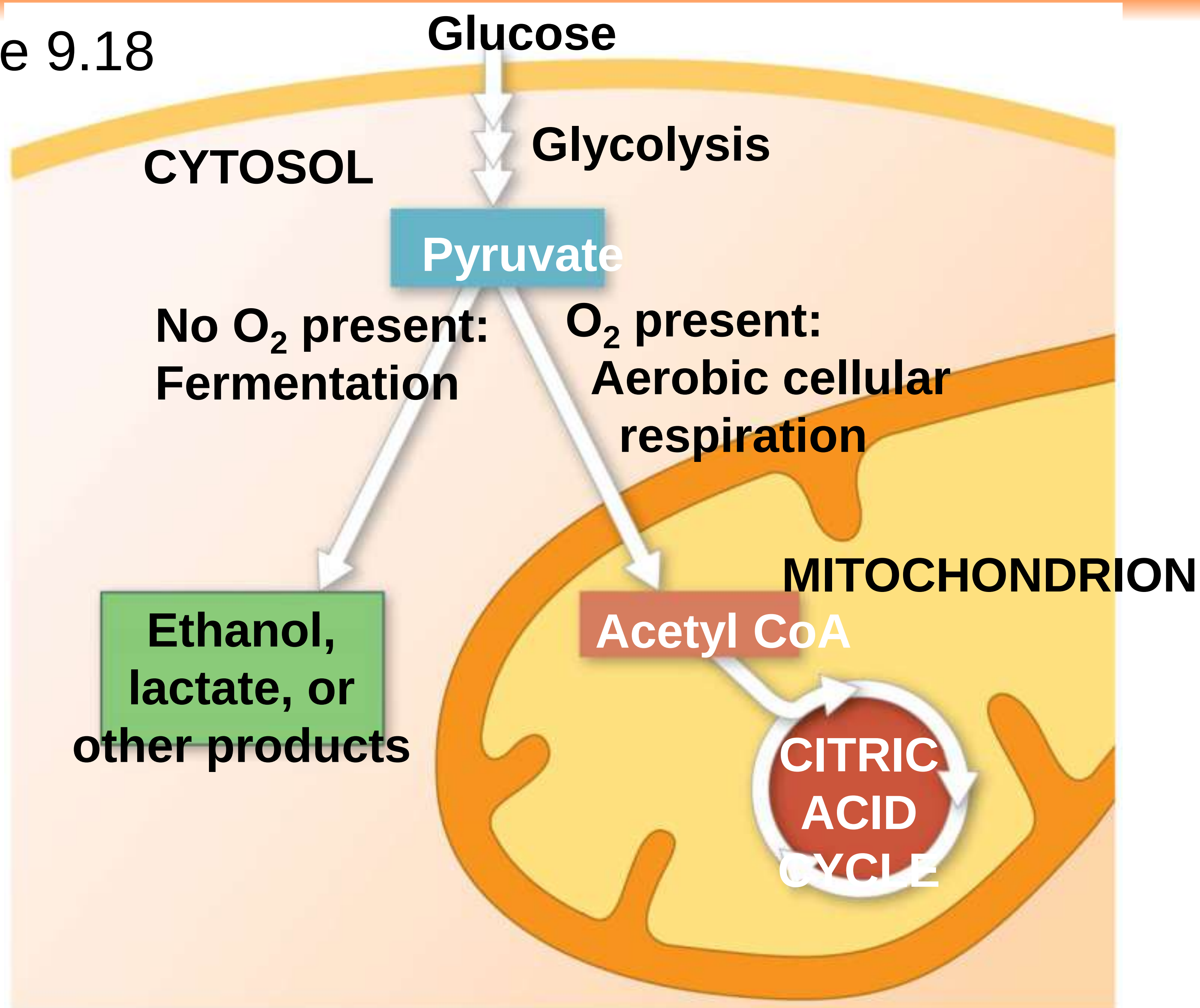
b) Cellular respiration produces 32 ATP per glucose molecule; fermentation produces 2 ATP per glucose molecule

a) **Obligate anaerobes** carry out fermentation or anaerobic respiration and cannot survive in the presence of O_2

b) Yeast and many bacteria are **facultative anaerobes**, meaning that they can survive using either fermentation or cellular respiration

c) In a facultative anaerobe, pyruvate is a fork in the metabolic road that leads to two alternative catabolic routes

Figure 9.18



Glucose
Glycolysis
CYTOSOL
Pyruvate
No O_2 present:
Fermentation
O_2 present:
Aerobic cellular
respiration
MITOCHONDRION
Ethanol,
lactate, or
other products
Acetyl CoA
CITRIC
ACID
CYCLE

The Evolutionary Significance of Glycolysis

a) Ancient prokaryotes are thought to have used glycolysis long before there was oxygen in the atmosphere

b) Very little O_2 was available in the atmosphere until about 2.7 billion years ago, so early prokaryotes likely used only glycolysis to generate ATP

c) Glycolysis is a very ancient process